WEB-BASED GIS PLATFORM FOR REAL-TIME LULC MONITORING

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CERTIFICATE

This is to certify that this project report titled **"Web-Based GIS Platform for Real-Time LULC Monitoring"** is a Bonafide record of work done by **CHENNARAO(228W1A1246) and BHAVANI PRASAD REDDY(228W1A1226)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**Web-Based GIS Platform for Real-Time LULC Monitoring**
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4	**Research Group**	**Deep Learning with Web GIS**
5	**Application Area**	**Environmental Sustainability**
6	**Aim of the Project**	**To develop a web-based GIS platform for automatic classification of satellite images into LULC categories using deep learning.**
7	**Project Outcomes**	• **Generates real-time LULC maps from uploaded satellite images.** • **Provides an interactive GIS dashboard for viewing and analysis.** • **Calculates and displays land distribution for each class.**

Student Signatures

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Signature of the Guide

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LIST OF EQUATIONS

Equation	BriefDescription
Eq (1)	Classification boundary
Eq (2)	Tree splitting decision
Eq (3)	Patch relationship learning
Eq(4)	Feature extraction model
Eq (5)	Efficient spatial learning

ABSTRACT

Land-use mapping and monitoring have become essential in today's world due to rapid urbanization and changing environments. However, even with the increasing importance of real-time LULC monitoring, traditional approaches to the analysis remain unchanged and employ inefficient techniques. They take time to perform manually and are not suitable for use in real-time systems because they do not have efficient ways to work with huge amounts of satellite images and cannot process information accurately enough. The present study is intended to develop a novel Web GIS that will provide real-time Land Use and Land Cover analysis through combining state-of-the-art techniques of deep learning and modern Web GIS platforms. The Web GIS will process satellite images using transformers like ViT and Swin-Transformer to classify lands into various types, e.g., urban, agricultural, water, or barren lands with high accuracy. This WebGIS will be able to monitor land uses using images taken with Landsat and Sentinel satellites. These processed datasets are then saved in GeoTIFF format and made available via web services hosted by GeoServer. The visualization process involves the use of an intuitive interface based on the Leaflet library, which provides functionalities like zooming, panning, and overlapping layers for exploring spatial data. Besides visualization, the software calculates area statistics for each land cover category and makes it available in both graphic and tabular form, making the analysis easier for end-users. Apart from this, the platform offers multi-temporal analysis capabilities, which enable users to analyze changes in land use over multiple periods. These can come in handy for detecting trends relating to urban sprawl, deforestation, and change in water resources. In terms of user experience, the software is very easy to use, as users will be able to perform the process of uploading satellite images and get instant results.

Keywords: Land Use Land Cover(LULC), Web GIS, Remote Sensing, Deep Learning, Vision Transformer(ViT), Swin Transformer, Satellite Imagery, GeoServer, PostGIS, Real-Time Monitoring

CHAPTER-1

INTRODUCTION

1.1 Origin of the Problem

The basis of this issue, which this project aims to solve, is related to the growing importance of continuously monitoring changes in Land Use & Land Cover (LULC). In recent years, the expansion of the urban landscape has happened at an extremely rapid pace [1, 2]. This resulted in many changes in the pattern of the land, which includes land being converted from agriculture to urban, decrease in water resources, and increase in barren lands. All of these factors affect [10] the sustainability of the environment, climatic conditions, and resource management [2].

The traditional approach to LULC analysis is dependent on manual interpretation of satellite data and offline processing tools. Even though it has been practiced for several years now, the traditional approach is both time-consuming and inefficient. This is because there are a lot of manual processes involved in analyzing satellite data, and it does not offer real-time analysis and visualization capabilities either [3]. The other main problem relates to the absence of integrated systems that would include the functionality of image classification, visualization, and analysis all in one place [4]. Currently, there are systems that either specialize in visualizing images or analyzing them; however, these functions are not implemented simultaneously with immediate feedback on results. Users cannot input images from satellites directly in order to get immediately classified outputs together with statistical data about their location and characteristics.

It should be mentioned that the use of classical ML algorithms or the pixel-based approach often faces difficulties [5] in achieving good results and scaling due to the increasing amount of satellite images. Given that the progress in DL, especially with transformer-based architectures [6], has been quite significant, there is a chance to solve this problem. That is why the purpose of the project is to create an automated, intelligent, and web-based system to monitor the LULC using satellite imagery with the help of deep learning.

Problem origin in LULC analysis is associated with the growing necessity to assess rapid changes that take place on Earth's surface due to natural and human-induced causes. Within the last several decades, such phenomena as urbanization, industrialization, population increase, deforestation, climate changes [7], and others contributed to a transformation of land patterns. Thus, there arose a need to create effective techniques that would facilitate the analysis of land use and help with sustainable development.

Previously, LULC mapping was conducted manually by interpreting cartographic materials and aerial photos. Despite some positive aspects of the method, it was rather difficult to work with and required substantial efforts. The development of satellite technology led to the widespread adoption of remote sensing methods that allowed for land cover analysis [8]. However, the first attempts at solving the problem were limited by simple techniques of pixel classification that were not able to cope with complicated landscapes. Another significant problem that helped in creating this problem is the lack of integration of classification systems and visualization systems [9]. There are a number of systems available that are primarily concerned with ensuring the accurate classification of data but lack visualization options that will aid users in analyzing results.

The rise in volume of geospatial data has created another set of challenges [10] for data analysis and visualization. It is becoming increasingly difficult to analyze vast amounts of satellite data in an effective manner using traditional machine learning algorithms. These algorithms may be effective, but they have trouble dealing with high-dimensional data and are unable to extract features automatically [11].

The rise in demand for real-time monitoring of data in different domains has created yet another challenge for existing systems. Current state-of-the-art solutions cannot handle real-time situations and are only suited for offline analysis purposes. The other cause of the problem can be attributed to the difficulty in discriminating different land cover types [12]. Agricultural land, barren land, and dry vegetation, for instance, might possess spectral features that are similar hence causing confusion. It calls for an elaborate and effective model that can incorporate spectral and spatial data.

With the recent advancements in the field of deep learning and web-based geographic information systems (GIS), it is possible to design an accurate and effective classification system. Nevertheless, efforts towards designing such a system are still inadequate. There seems to be the lack of integration between accurate land use classification, efficient processing power, and an interactive visualization process [13].

From the above description, it is evident that the root of the problem is inherent in the existing traditional techniques. In addition, the advancement in the nature of geospatial data has made it impossible to develop efficient land use and cover systems.

1.2 Basic definitions and Background

The Land Use Land Cover (LULC) classification is the process of categorizing the earth's surface [14] into urban, agriculture, water, forest, and barren lands. LULC is an important topic in the study of the environment because it enables us to comprehend how we are utilizing our environment and how it may be changing over time. LULC data are crucial in urban planning, resource management, and environmental protection efforts. Remote sensing is the acquisition of information about the Earth's surface using satellite images without physical contact [15]. Satellites like Landsat and Sentinel take multispectral images of the Earth's surface that enable identification of various land cover types depending on their spectral signatures. Remote sensing data are popularly used in LULC mapping because they offer frequent observations of wide coverage [16].

Deep learning is a new generation of artificial intelligence applications that have been implemented in image classification tasks [17]. Advanced deep learning models such as the Vision Transformer (ViT) and Swin Transformers are used in this research. Attention-based architectures in vision transformers improve classification accuracy by extracting long-range dependencies and hierarchical interactions in satellite images. Web GIS is a technology where the users can access, analyze, and visualize the spatial data with the help of the browser [18]. Web GIS uses the functionalities of GIS along with the features of web technology to create a platform for interactive maps. For this project, software such as GeoServer is used to serve the spatial data, and Leaflet library is used to display the maps in the web environment.

PostGIS is an extension of the PostgreSQL database management system that helps manage and store the spatial data in the system [19]. PostGIS is being used in our project for storing the classified data and the area information. GeoServer helps act as an intermediary between the database and the frontend by offering services such as WMS.In this project, remote sensing technology has been integrated with deep learning and Web GIS technology to make a fully integrated system for performing LULC classification.

Land Use and Land Cover (LULC) are interlinked yet distinctly different notions [20]. Land cover implies the presence of any material on the surface of the earth, such as vegetation, water, soil, or constructed areas. Land use can be defined as the way people interact with the surface of the earth, which may include activities like farming, housing developments, industrial zones, or leisure purposes. Thus, the distinction between LULC and land cover should be considered when interpreting satellite imagery.

Satellite remote sensing implies gathering data about objects on the earth's surface using non-contact methods like sensors aboard satellites and aircraft [21]. Remote sensing offers great advantage in monitoring changes in land cover due to its wide range of application and regular acquisition. Satellite imagery consists of several spectral bands, which allows distinguishing various land elements based on their spectral characteristics.

A spectral signature is a set of electromagnetic radiation absorption or reflection values specific to certain substances. Thus, water bodies have low values of reflectance and absorb almost all types of radiation; vegetation reflects light predominantly in near-infrared band. Spectral signatures are used by classification algorithms to differentiate land covers [22].

Image classification involves mapping each pixel of a satellite image to its respective land cover category [23]. The two main approaches in carrying out image classification include supervised and unsupervised image classification. Supervised image classification requires labeled data for training, while unsupervised image classification involves grouping similar pixels into distinct classes without using labeled data.

Spatial resolution is another key concept in remote sensing. This term relates to the smallest size of the feature that can be observed on a satellite image. Images having high spatial resolutions have more details but need more space and computational capacity to store. Temporal resolution involves capturing images at certain frequencies.

Geographic Information System (GIS) is another important concept in remote sensing. This is the system that enables one to manage and analyze geographical information [24]. It involves creating a database that integrates geographical data in order to make geographical analysis easy. The development of Web GIS has revolutionized conventional GIS technology in allowing online access to geospatial data [25]. The use of Web GIS makes map interaction easier since one can visualize maps in their browser. It offers a wide array of benefits including real-time sharing of data and easy visualization and analysis of data.

Deep learning and machine learning are other important concepts in the context of LULC classification [26]. The former involves handcrafted features while machine learning algorithms such as SVM and Random Forest use deep learning models to extract features. Change detection is another crucial area of concern in land cover analysis. It can be used to identify changes like urbanization, deforestation, etc [27].

Lastly, the combination of geospatial data and artificial intelligence lays the groundwork for modern LULC systems. Modern satellite imagery, combined with machine learning and Web GIS technologies can lead to the creation of smart systems that would perform land use analysis.

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

LULC analysis plays a key role in understanding environmental changes, urban growth, and resource allocation. Nonetheless, current LULC analysis tools and processes mainly rely on manual procedures which are time-consuming, and expertise in handling satellite data is required. These conventional techniques include several stages ranging from data acquisition and preprocessing, classification, and visualization using various tools that lead to an inefficient process.

The second constraint is that existing solutions do not offer real-time and interactive features. There is no integrated platform that allows users to upload satellite images and receive instant results of LULC maps along with analytical features. The current software only offers outputs without any interactive options such as zooming, controlling layers, and overlays on basemaps.Besides, the current conventional classification algorithms have poor performance in analyzing huge volumes of satellite images. Conventional classifiers cannot effectively recognize spatial variation and patterns in land cover attributes. Thus, there is a requirement for more advanced methods based on deep learning.

Therefore, it is necessary to create a fully automated, reliable, and easy-to-use web application for analyzing satellite images.

1.3.2 Objectives

The primary aim of this project is to develop a web-based GIS application that will be able to monitor the Land Use and Land Cover using Deep Learning techniques and geospatial technologies in real time. The tool will have features like satellite image classification and visualization for the purpose of understanding land distribution and taking necessary decisions.

The detailed objectives of this project are as follows:

1. Develop Deep Learning algorithm for automatic satellite image classification.The system should have an ability to properly classify the satellite images based on the type of land cover which includes urban areas, agriculture lands, water bodies, and barren lands among others. For this purpose, sophisticated techniques like ViT and Swin Transformer are used. The objective of this goal involves improving the process of classification and handling complicated satellite imagery.

2. Develop a web interface for uploading images and result visualization.It is important to develop a web interface that will allow users to upload satellite images and visualize their classification. The system will show the classification results with proper color codes which makes it easier for the users to understand the land cover classification map.

3.To publish the classified LULC using GeoServer as web services.The classified LULC is published through GeoServer in the form of web services such as WMS and WFS. In this way, seamless integration between the backend and frontend becomes possible and real-time map visualizations are achieved. This will help to handle and share the geospatial data properly in a standard manner.

4. To compute and visualize area statistics of land classes automatically.The area statistics of various land classes are calculated and displayed in a systematic form. It will enable users to analyze land use patterns effectively. In this way, the analytical capabilities of the proposed system are strengthened in addition to the visualization.

5. To develop an interactive and scalable Web GIS application for decision support.The last goal of the project is to design a complete system which provides interactive mapping functionality such as panning, zooming and overlapping maps. The proposed application should be scalable for future use in big datasets and real-time monitoring.

1.3.3 Outcomes of the Project

The outcomes achieved by this research project showcase the successful integration of a fully operational Web-Based GIS tool that can be used to monitor LULC in real-time through automatic classification of the provided images and generation of LULC maps.

One of the important achievements that came out of the implementation of the project was the successful integration of image processing and classification using deep learning techniques in order to enable users to upload satellite images and get their classified versions immediately.

Another key achievement is the success in developing an effective GIS interface that will allow users to navigate the map and perform actions such as zooming, panning, and overlay of the layers in order to visualize the LULC maps effectively.

Also, one of the important features implemented in the project is the provision of automatic calculation and visualization of class-wise land distribution, which helps users understand how much each type of land occupies.

From a technical perspective, the integration of GeoServer and PostGIS ensures efficient storage, management, and sharing of spatial data. The use of advanced deep learning models improves classification accuracy and performance when working with complex satellite imagery.

Overall, the project delivers a complete and scalable solution that combines automated classification, real-time visualization, and spatial analysis. It reduces manual effort, improves efficiency, and provides valuable support for applications such as urban planning, environmental monitoring, and land management.

1.4 Related Applications

There are many applications of the Web-Based GIS Platform for Real-Time LULC Monitoring in different sectors of society.

One prominent field is urban planning. Urban planners can employ the web-based GIS system to monitor any urban sprawl and control any change that may occur regarding the land usage. They can plan the infrastructure in cities on the basis of data provided by the system. This system is also quite beneficial in environmental monitoring as well. Through this system, one can track any change in the forest cover, water bodies, and barren lands. Such data can help in the identification of issues like deforestation, land degradation, and depletion of water resources.

Lastly, it is very useful in agricultural monitoring too. Farmers can utilize the data of the system for better crop planning and agricultural management practices. Apart from that, the framework will help in managing disasters. It can be used to measure the effects of natural disasters like floods, droughts, and cyclones through multi-temporal image comparison. Such an approach will allow the authorities to formulate strategies to help in the recovery process.

The other important application of the proposed framework is the development of smart cities. It is because monitoring of land in real time will assist in optimizing resource use. Generally, the project presents a flexible and intelligent framework that could be employed across different areas.

Another significant area where the system could be used is climate research. It can aid in tracking changes in land cover and the impacts of these changes on various climatic factors including temperature, rainfall, and greenhouse gas emission levels. Monitoring of deforestation, urbanization, and decreased green coverage could contribute to environmental research and help devise methods that would allow to reduce risks connected with climate change. The platform could be helpful in the field of water resources management too. It would help find out where water reservoirs like lakes, rivers, and reservoirs are located, and track any changes that occur in them. Such data could prove valuable for managing water resources and creating irrigation infrastructure in case of water shortages.

Finally, the system could be utilized to protect biological diversity. Through analyzing forests and plant cover changes, the system could assist in identifying biotopes and monitoring ecosystem changes. It would be helpful for nature protection and preserving the natural environment from degradation caused by human activity. In the transportation and infrastructure sector, the model can be utilized in the evaluation of land suitability for road construction, railway development, and other infrastructure projects. The system assists in identifying the most suitable location for such developments without compromising the environment.

Disaster risk assessment and mitigation are some of the areas that benefit from the utilization of the system in monitoring land use and cover changes. In this regard, the model identifies potential areas that are susceptible to flooding, landslide, and drought incidences. This is essential in the formulation of strategies to minimize the risks of disasters. In the exploration of minerals, the system can be employed in evaluating land disturbance due to mining activities. This facilitates the measurement of the impacts of mining on the environment and formulating strategies for mitigation. Management of coastal zones is another sector that benefits from the utilization of the platform in monitoring changes in the coastal land cover.

A fourth application area for the system involves the use of it in smart agriculture to assist farmers in analyzing land suitability, crop patterns, and water needs in relation to irrigation. The system allows users to make decisions in precision farming. Another

way that the system can be utilized is in governance and decision-making by government institutions using LULC to develop land use policies, zoning systems, and sustainable development strategies. This will give an informed perspective regarding decision making in such areas.

Lastly, the system may also be applied to educational and research activities to aid students in their studies in land cover analysis and geospatial data among others. This would enable students to learn how remote sensing works.

CHAPTER-2

REVIEW OF LITERATURE

2.1 Description of Existing Systems

An early LULC classification system based on Support Vector Machines (SVM) was presented in, where spectral features from satellite imagery were used for land classification [1]. The study demonstrated that SVM provides better accuracy compared to traditional classifiers, but it still struggles to capture complex spatial patterns in large-scale datasets. A pixel-based classification approach using Maximum Likelihood Classification (MLC) was discussed in [2]. This method relies purely on spectral information for classification and is widely used in remote sensing. However, it lacks the ability to consider spatial relationships between pixels, which limits its performance in complex environments.

An object-based image analysis (OBIA) system was introduced in, where both spectral and spatial features were considered for classification [3]. The system improved accuracy compared to pixel-based methods, but it required complex parameter tuning and higher computational resources. A change detection system for multi-temporal satellite images was presented in, focusing on identifying land changes over time [4]. The study highlighted the importance of temporal analysis but did not integrate real-time visualization or automated classification.

A machine learning-based LULC classification system using Random Forest was proposed in [5]. The system improved classification performance by learning from training data, but it required manual feature extraction and was not fully automated. A k-NN based classification approach was discussed in, which classified land cover based on similarity measures [6]. Although simple and easy to implement, the method showed limitations in handling large datasets and complex land patterns. A deep learning-based LULC classification system using Convolutional Neural Networks (CNNs) was presented in [7]. The study showed that CNNs can automatically extract features and improve classification accuracy compared to traditional methods.

An advanced CNN-based framework for satellite image classification was introduced in [8], demonstrating improved performance in handling high-resolution imagery. However, the system lacked integration with GIS platforms for visualization. A transformer-based model called Vision Transformer (ViT) for image classification was proposed in [9]. The study demonstrated that attention mechanisms can capture global features effectively, improving classification results in remote sensing applications. A hierarchical transformer model known as Swin Transformer was presented in [10], which improved spatial feature extraction using shifted window attention. This model showed better performance in large-scale image analysis tasks. A multi-source data integration system combining optical and SAR data was discussed in [11]. The approach improved classification accuracy but increased system complexity and computational requirements.

A data fusion-based LULC classification approach was proposed in [12], where multiple datasets were combined to enhance classification results. However, it required extensive preprocessing and was not suitable for real-time applications. A Web GIS-based system for spatial data visualization was introduced in [13], focusing on map rendering and user interaction. The system enabled online visualization but did not include automated classification features. A web-based geospatial platform integrating GIS services was presented in [14], which allowed data sharing and visualization. However, it lacked deep learning integration for automated analysis.

A deep learning-based LULC classification system integrated with Web GIS was proposed in [15], highlighting the importance of combining AI and GIS for real-time applications. This approach showed improved usability and efficiency but still required further enhancement for full automation. The work presented in [16] investigated the application of Decision Tree classifier in LULC map generation. It uses decision tree rules for land cover type classification using spectral and spatial characteristics. Even though the model was straightforward and interpretable, it was inefficient in dealing with complicated data sets, and overfitting would occur easily when the data variation was high.

Another model proposed in [17] is k-Nearest Neighbors (k-NN) classifier that classifies each pixel using similarity measure from its neighbor samples. While it is easy to apply

and works effectively with small data sets, it is not scalable to large data sets and satellite images and is prone to noise interference. In [18], the authors proposed a system that integrates SVM with feature extraction techniques. The proposed system enhances the accuracy of classification through the selection of significant features in the satellite image dataset. But it still requires intensive human effort for feature engineering and model tuning.

Lastly, the work described in [19] investigated the use of ANN in LULC classification. Compared to classical machine learning algorithms, ANN can learn nonlinear patterns of data and outperforms other models in classification performance. However, it needs more training samples and has higher computational cost during training. Another system from [20] considered object-based image analysis (OBIA), whereby an image is divided into meaningful objects prior to classification. OBIA provided better spatial accuracy than pixel-based methods but involved a more complicated segmentation process with the need for numerous parameters and high computation costs.

An ensemble learning technique was applied to create a land use and cover analysis system in [21]. Higher accuracy and stability were achieved at the expense of more complex system architecture. A system for conducting LULC analysis in a cloud environment was suggested in [22]. The platform used Google Earth Engine to conduct LULC analyses involving huge sets of satellite images. Such a system provides efficient data processing. However, the internet connection is necessary, and there are some drawbacks related to limited flexibility. Time-series satellite data can be used to perform change detection and classification in [23]. Although this technique provides more accurate results, it involves higher computational costs and the availability of big amounts of data.

Deep learning models, such as the U-Net model, have been used for LULC classification as in [24]. They delivered pixel-level accuracies and detailed output, but their operations involved complex computations and huge amounts of annotated data. The proposed solution in [25] incorporated GIS along with machine learning models for spatial analysis. It offered visualization and classification but did not provide real-time processing support and lacked advanced deep learning support. Moreover, in [26], researchers investigated transfer learning algorithms for LULC classification that

reused pre-trained models to reduce learning time and improve accuracy. In addition, transfer learning relied on availability of appropriate pre-trained models.

Multi-resolution analysis has also been studied for LULC classification in [27]. Multi-resolution analysis combines images acquired at different resolution levels to achieve better accuracy of classification. Nonetheless, the analysis is computationally intensive. Fuzzy logic based classification techniques have been introduced in [28] to handle uncertainty in land cover classification. The application of fuzzy classification improved classification results in doubtful areas.

2.2 Summary of Literature Study

From the literature review, one can see how the LULC classification has changed over time. At first, researchers started developing LULC models using image processing and classification algorithms. For example, [1] showed that the Support Vector Machines and Random Forest algorithms could be used to classify satellite images with reasonable accuracy. Nevertheless, their performance was heavily dependent on manually extracted features. Later, studies started introducing machine learning and deep learning algorithms to solve problems of satellite imagery processing. It became evident that deep learning architectures were much more efficient in terms of scalability and classification accuracy. The best results were achieved through Convolutional Neural Networks (CNN). Studies showed that CNNs were much more efficient than other architectures. Moreover, CNNs' performance could be further improved by combining multiple algorithms.

One of the main trends in the current literature is the use of transformers in image classification. Vision Transformer (ViT) and Swin Transformer architectures have proven to be much more efficient than the previously proposed architectures for remote sensing data classification. They use an attention mechanism to extract both local and global features from images. Therefore, they have excellent potential for classifying complex images.

Moreover, the research literature shows the tendency to use multi-source integration to enhance the effectiveness of classification techniques through the combination of optical, SAR, and LiDAR datasets. This technique allows overcoming such drawbacks

as cloud cover presence and data incompleteness but makes the system more complex and requires greater computational resources. Furthermore, the recent literature discusses the application of Web GIS technologies to perform real-time visualization of data. Using such systems, users can work directly with maps, conduct data analysis, and make decisions. Nevertheless, most systems utilizing Web GIS are oriented exclusively at visualization; therefore, no classification procedures are implemented.

Notwithstanding numerous achievements, most of the systems considered in the literature pay attention to some particular aspects of the problem rather than to the whole picture. Moreover, most systems are oriented exclusively at classification or visualization; thus, there are no systems providing an integrated approach for both purposes. Besides, all of them operate in an offline environment and lack real-time processing capacity [30]. In general, the review shows that despite the progress that has been achieved through the use of deep learning and remote sensing technologies for the purpose of classifying LULC, there is still room for creating an integrated system capable of performing automatic classification, processing data in real-time, and visualizing the results interactively. This is what the proposed Web-based GIS application aims to achieve.

CHAPTER-3

PROPOSED METHOD

3.1 Design Methodology

The Design Methodology of the Proposed System .This section discusses the design methodlogy used in designing the system architecture. As mentioned earlier, the aim of the proposed system is to create a framework capable of automated Land Use Land Cover classification and web-based visualization using a Web GIS technology. Thus, unlike previous LULC applications, in which image processing, classification and visualization have been separate processes, our methodology aims at addressing the limitations of existing techniques by combining geospatial techniques with deep learning models. In other words, satellite images would be processed, classified and displayed through one process only.

Data Collection and Preparation In order to perform effective classification of land use and land cover, satellite imagery data must be first acquired. In this regard, Landsat and Sentinel satellites will be used to collect the required imagery data. Images provided by these satellites include multspectral images in GeoTIFF format. Furthermore, the data collected from these satellites include valuable information such as urban area, vegetation, waterbodies, and barren land.

Following data collection, the next phase would be data preprocessing. The satellite images would need to undergo processing to guarantee that they will be consistent and of high quality to serve as inputs to the machine learning algorithm. Data preprocessing entails procedures such as selecting specific bands, normalizing the images, changing their dimensions, and extracting patches. For instance, the current project would entail dividing images into patches, which could be 64×64 in size, to suit deep learning algorithms.

After data preprocessing, the next phase would involve training the model. In this phase, complex deep learning models, such as Vision Transformer and Swin Transformer, would be used to train the dataset. These models utilize attention mechanisms to extract global

and local features from images. Model training would require feeding the model with labeled data and optimizing its performance. After training, the algorithm conducts the classification and prediction tasks. In this case, when an individual uploads a particular satellite image, it is put through the trained algorithm, which then classifies the images into different LULC categories like urban, agricultural, water, and bare lands. Combining the outputs of these classifications creates a classified map from satellite imagery.

Further, the methodology continues with the creation of a GeoTIFF file and spatial data management. In this process, the outputs of the classified image are made into a geo-referenced TIFF file in order to allow for further manipulation in GIS software. Next comes the process of storing the database in PostGIS. The classified results and area statistics are stored in the PostgreSQL database having PostGIS functionality. It will allow easy storage and manipulation of spatial data. Also, it provides the calculation of area statistics for each land cover class.

After this, area statistics are calculated where the area coverage of each class (urban, water, agricultural, and others) is calculated. The area statistics are converted into percentages which are shown to the user. The last step of the methodology involves the visualization and interaction with the user. A frontend user interface is created using HTML, CSS, JS, and Leaflet.js. The user will be able to upload the satellite images and perform further interactions like classification and analysis of the result. System-wise, the whole methodology may be considered as a pipeline comprising data acquisition, pre-processing, training model, classification, integration with GeoServer, storing in the database, and visualization. Every step interconnects with others, thus providing a full-fledged end-to-end system.

In conclusion, one should say that the methodology under discussion offers an effective approach to real-time monitoring based on LULC. Being based on both deep learning and Web GIS technologies, the system not only facilitates automatic classification, but also makes the process much more efficient and comfortable from the standpoint of visualization.

3.2 System Architecture Diagram

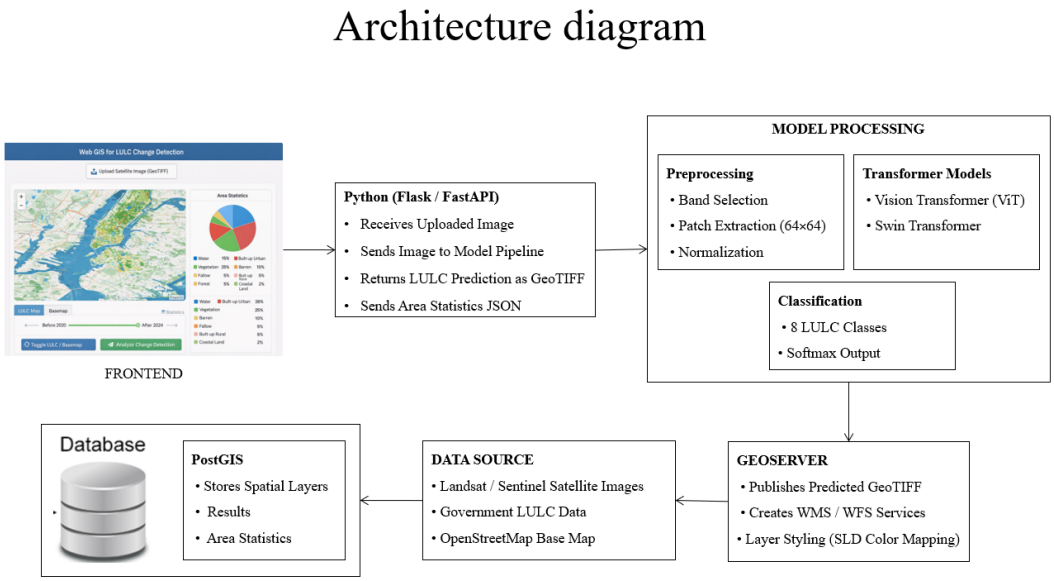

Fig 3.1: Architecture Diagram

The architecture of the proposed Web-Based GIS Platform for Real-Time LULC Monitoring System involves a combination of deep learning-based image classification with Web GIS visualization and spatial database management in a holistic pipeline. The key objective behind the system's architecture is that apart from being able to classify satellite images according to their LULC, it also needs to provide real-time visualization and analytics through an intuitive user interface. This means that the system will not follow the traditional approach whereby the image classification task is performed independently of the visualization process.

Fig 3.1 shows the different elements of the architecture of the system, including the Frontend, Backend (Python), Image Classification Models, Model Processing Unit, GeoServer, PostGIS Database, and Data Source. Firstly, the workflow starts from the frontend. The user uploads the image to the system, which is done in GeoTIFF format. Apart from this, there are other frontend components such as map visualizations, zooming, panning, and area statistics. Hence, as per Figure 3.1, the frontend forms the interactive layer between the user and the system. After uploading, the image goes to the backend

server, which uses Python programming language to perform the tasks. These operations include the reception and transmission of images by the server to the processing model. Further, after processing, the server delivers back the classified LULC image and the area statistics to the user. Thus, as per Figure 3.1, the backend acts as the medium of communication between the UI and the processing module.

Finally, the last component is the model processing unit. As can be seen in Figure 3.1 below, there are three modules in the model processing: pre-processing, transformers model, and classification. These three modules carry out specific operations to classify land use. Firstly, the operations done in the pre-processing stage include band selection, patch creation (64×64), and normalization of input data. Once the preprocessing step is completed, ViT and Swin Transformers are employed by the system to extract features and perform classification. The models can extract features on both the global level and the local level. As a result, each patch in the input image will be assigned to a certain LULC category, which includes categories such as urban, agricultural, water, and barren lands. Softmax is applied to produce the probability distribution of each class. Hence, the model processing unit converts satellite images into LULC outputs, as shown in Figure 3.1.

PostGIS database is also part of the system, storing spatial layers, classifications, and area information. The PostGIS database guarantees efficient data storage and access, as seen in Figure 3.1 below. Other inputs for the system include data from sources such as Landsat and Sentinel satellites' images, government LULC, and OpenStreetMaps base maps. In terms of the overall system workflow, as depicted in Figure 3.1 above, the architecture works on the principle of the following process: image upload → processing → classification through deep learning model → publishing through GeoServer → storage in database → visualization.

In summary, the architecture, shown above in Figure 3.1, is an efficient solution that allows real-time LULC monitoring in real time. Through the integration of deep learning models and Web GIS technology, not only is it easy to automate the classification process but also enhance visualization.

3.3 Description of Algorithms

3.3.1 Classification of SVM and RF

1. Support Vector Machine (SVM):

SVM is indeed a powerful, widely used supervised learning algorithm for LULC classification because of its capability to handle high-dimensional and nonlinear data. In this study, SVM will be used to classify different land cover types based on the spectral characteristics of satellite imagery. The algorithm works by identifying an optimal hyperplane that separates various classes with the maximum possible margin. The points which lie closest to the hyperplane are called support vectors, which define the decision boundary. When data are not linearly separable, kernel functions such as linear, polynomial, or RBF are applied to project the data into a higher-dimensional space where separation becomes possible.

In the n-dimensional feature space, the separating hyperplane can be mathematically defined as:

$$w^T x + b = 0 \qquad \text{Eq (1)}$$

Where,

w is the weight vector that is perpendicular to the hyperplane.

x represents the feature vector, or input data, and

b is the bias term, which shifts the hyperplane from the origin.

SVM was used in this study as a supervised classifier, utilizing labeled training samples to discriminate among multiple LULC categories. Its ability to create clear and well-defined decision boundaries makes it suitable for complex satellite imagery classification. Support Vector Machines (SVMs) have been proven to work effectively with high-dimensional remote sensing data, which consists of several spectral bands for each pixel. This algorithm is thus ideal for satellite image classification where feature values are often very complex. Firstly, the biggest advantage of SVMs lies in the selection of support vectors alone as opposed to working with all available data points. It is through this process that SVMs manage to perform much more efficiently and accurately than algorithms using all dataset

points. Secondly, an SVM can deal very well with the nonlinearity in the relationship between different types of land cover. Real world examples include areas of agricultural lands, vegetation, or barren lands that have similar spectra. Through the use of kernels like the Radial Basis Function (RBF), an SVM algorithm is able to transform such data into a multidimensional space where separation can be done much more easily.

Moreover, it is evident that SVM has a good generalization property, which implies that it works not only during training data processing but also during testing data analysis. This feature becomes critical when considering SVM applications in LULC classification because here the algorithm should be able to perform correctly regardless of whether it classifies objects from a known or unknown region or period. On the other hand, there are several drawbacks of using SVM on large-scale data sets. First, its training may turn out to be computationally heavy when too many samples are provided. Second, choosing proper kernel function types and adjusting the parameters C and gamma requires experimenting. Nevertheless, despite all mentioned above, SVM is still considered to be one of the most accurate algorithms for remote sensing applications.

2. Random Forest:

Random Forest is an ensemble-based machine learning algorithm that works by constructing numerous decision trees in the course of training and then combines outputs for final classification. Each tree is grown on a random subset of training data and features, hence assuring diversity within the model itself and reducing overfitting. In the context of LULC classification, RF skillfully manages large datasets with mixed spectral information along with noise, making it highly robust for applications in remote sensing. The basic concept of RF is to create N decision trees. Each tree will give one classification result, and the final class will be determined by the majority vote from all the trees. The quality of each tree's split is measured using metrics such as Gini impurity or entropy, defined as:

$$G = 1 - \sum_{i=1}^{n} p_i^2 \qquad \text{Eq(2)}$$

Where,

p_i is the probability of samples belonging to class i.

Being an ensemble technique, RF improves generalization and accuracy, especially in the discrimination of spectrally similar classes such as vegetation, forest, and agricultural regions. High tolerance to noise and handling complex and nonlinear relationships make the model quite reliable for large-scale LULC mapping and monitoring applications. There are a number of benefits that are associated with the use of Random Forest algorithm for LULC mapping from satellite imagery. First, this algorithm works very well with high dimensional data, and there is no need for feature selection or feature scaling. The reason behind this is the fact that RF selects at random a certain number of features when constructing a decision tree, hence, identifying automatically the best features for classification.

Secondly, Random Forest algorithm is not very sensitive to noise because of averaging of the output of different decision trees. This is because the probability of obtaining the same wrong result in all of them is minimal. Therefore, RF can work very well with noisy satellite imagery. Random Forest can also generate feature importance metrics. Feature importance metrics will help identify the contribution of each input variable to the classifying process. In our case, we can learn how spectral bands or indices affect the distinction between various land cover classes.

Moreover, RF is a powerful method to classify datasets with unbalanced classes. In other words, the number of samples for some classes can be significantly higher than for others. However, the use of bootstrapping and sampling techniques allows RF to generate balanced subsets during training, which increases the accuracy of the classification algorithm. However, there is one drawback associated with RF. Classification results generated by this algorithm might not be as smooth as those created by deep learning algorithms because it classifies images pixel-by-pixel. Despite this, Random Forest remains an effective algorithm for LULC classification because of its simplicity and accuracy.

3.3.2 Classification of ViT and Swin Transformers

The suggested framework employs state-of-the-art deep learning algorithms that facilitate precise LULC classifications. The primary algorithms employed in the current research are Vision Transformer (ViT) and Swin Transformer. The mentioned models are intended to process complicated satellite images through identifying global and local spatial relationships within them. ViT and Swin Transformer algorithms are superior to machine learning and CNN-based models.

1. Vision Transformer (ViT):

In contrast to other architectures, the ViT uses a method of breaking down the input image into patches of a small fixed size as opposed to working on the image as one unit. These patches are considered tokens and are projected into numbers based on linear projection. Because transformers do not have any intrinsic knowledge of spatial relationships, position embeddings are assigned to each token to help preserve their spatial relationship. These tokens are passed to several layers of transformers where they are analyzed for relationships. In each transformer layer, the image tokens undergo a multi-head self-attention mechanism followed by a feed-forward neural network to help refine the features extracted. Through self-attention, the image can be understood globally as the model is able to extract the dependencies of each token across all of them, giving it an insight into how each image part relates to others. Finally, the output from the transformer layers is passed through a classifier layer, which generates probabilities for different class outputs.

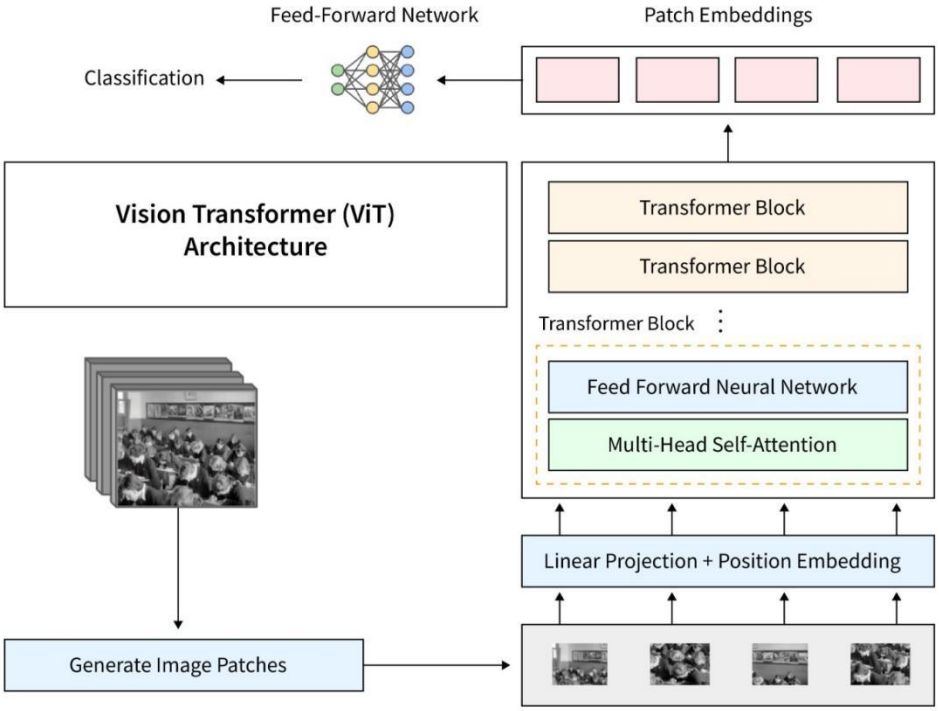

Fig 3.2: Vision Transformer Architecture

Vision Transformer (ViT) is a deep learning-based algorithm that utilizes the transformer model, which was developed initially for natural language processing tasks. Instead of analyzing the whole image at once, ViT breaks down the image into small patches of size 64×64. Then each patch becomes a token, and then it undergoes through the transformer algorithm.

$$z_i = Ex_i + p_i \qquad \text{Eq (3)}$$

ViT uses a self-attention function, enabling the model to comprehend relationships within the patches. In this way, it captures global features, which is a key aspect in case of satellite imagery as the features are distributed widely. The algorithm analyzes all the patches concurrently and learns their dependencies. In the present study, ViT will be applied to patches of satellite images and classify them into different LULC categories including urban area, agriculture land, water bodies, and barren lands.

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V \qquad \text{Eq (4)}$$

Where,

- Q: *Query matrix*
- K: *Key matrix*
- V: *Value matrix*
- d_k: *Dimensionality of the keys (scaling factor)*
- QK^T: *Similarity score between patches*
- *Softmax: Converts scores into weights*
- *Output: Weighted combination of patch features*

2. Swin Transformer:

The swin transformer expands on the traditional transformer by incorporating the concept of hierarchy and shift window attention, which allows the effective handling of large-scale satellite imagery. Rather than using global attention across the entire image, the image is divided into smaller windows and attention calculations are performed within these windows. The attention within a window Wwith patches x_1,x_2,…,x_nis calculated through the use of a scaled dot product attention formula, however limited to those patches that reside within W.

By shifting the window attention as per Eq[3], the model can exploit both local and global features of the image. Local features help capture fine details of the LULC boundaries, while global features help differentiate between class types such as agricultural and fallow lands. During training, patch merging results in decreased spatial resolution and larger channel dimensions, thus creating a hierarchical architecture that mimics a CNN backbone with transformer capability.

$$\text{Attention}(Q_W, K_W, V_W) = \text{softmax}\left(\frac{Q_W K_W^\top}{\sqrt{d_k}} + B\right) V_W \qquad \text{Eq[5]}$$

Where,

- B: *Relative position bias, which provides spatial awareness inside each window*

- *Q: Query matrix*
- *K: Key matrix*
- *V: Value matrix*
- *dk: Dimensionality of the keys (scaling factor)*
- *QK^T: Similarity score between patches*
- *Softmax: Converts scores into weights*
- *Output: Weighted combination of patch features*

Introduction The Swin Transformer is a sophisticated model based on Vision Transformer architecture. Like its predecessor, the Swin model starts with patch partitioning and embedding. Nevertheless, unlike ViT, the model uses attention computations not globally but locally within the non-overlapping windows, making calculations much easier. In addition, the Swin Transformer implements a shifted window mechanism, meaning that the model shifts the windows between each of its layers. As a result, the Swin Transformer can build dependencies among nearby parts of the input. The Hierarchical Structure of the Model The Swin Transformer utilizes a hierarchical architecture in which the spatial resolution of feature maps is gradually reduced while increasing the channel size. Each stage comprises several Swin Transformer blocks performing self-attention computations within the windows and using a feed-forward network. In addition, there are shifted windows at the second and following stages to make sure the model creates links among close parts of feature maps. In such a way, the architecture helps improve the efficiency of the algorithm and make the latter applicable to the processing of images with high resolutions. Finally, the model passes feature vectors through the classification layer.

3.4 Description of datasets and Tools

Web-Based GIS Platform for Real-Time LULC Monitoring System is built using a blend of satellite data and geospatial, and deep learning methods. Data and methods mentioned here have been found to be critical in facilitating accurate classification, efficient processing, and visualization of land use patterns.

1. Datasets Used

Satellite imagery datasets are mainly used for land use classification purposes. The two datasets used in this case include:

a) Landsat 8 Satellite Imagery

Landsat 8 provides multispectral images with moderate spatial resolution. This dataset is extensively used in environmental monitoring and LULC studies. Several bands are included in this dataset, including visible, infrared, and thermal bands. Bands in the dataset make it easy to detect different land features such as vegetation, water, and urban areas.

b) Sentinel-2 Satellite Imagery

This is another satellite imagery dataset that provides high-resolution optical images. It is known for its improved spatial and temporal resolution compared to other datasets such as Landsat 8.

c) GeoTIFF Format

All satellite images are processed in GeoTIFF format. This particular format has the property of storing images together with other geographic data such as coordinates and projections. The GeoTIFF format is very necessary for merging the results obtained to GIS systems.

d) Ground Truth Data

Ground truth data is required for training and validation of the proposed deep learning algorithm. This data consists of labels for different types of land use, such as urban, agricultural, water, and barren lands. The ground truth data teaches the algorithm to recognize the patterns.

e) Multi-Temporal Data

Multi-temporal satellite imagery data refers to the process of acquiring satellite imagery data in different periods of time. It assists in detecting the changes in land uses, including urbanization, deforestation, etc.

2. Tools & Technologies Involved

There have been many tools and technologies involved to create an all-in-one system.

a) Python (Flask/FastAPI)

Python serves as the backend programming language. Tools such as Flask/FastAPI are used for uploading images, handling requests, and interfacing between the front end and the deep learning model. This works as the processing engine of the entire system.

b) PyTorch

PyTorch is used for creating and training deep learning models like Vision Transformer (ViT) and Swin Transformer.

c) PostgreSQL with PostGIS

PostgreSQL is utilized as the database and PostGIS is the spatial database extension that will be used to store spatially classified data, processing results, and statistics about the analyzed area.

d) Web Development Technologies (HTML, CSS, JavaScript, Leaflet.js)

The frontend is built using web programming languages and Leaflet.js is used for mapping visualization purposes. This will provide the opportunity to interactively display map layers on which one can pan and zoom around.

e) Google Earth Engine

This tool will be used for downloading satellite image datasets.

3. Dataset Processing and Visualization

Incorporating satellite images and software for analysis will mean loading satellite images into preprocessing tools and sending the processed images to the deep learning models for analysis. Afterwards, the classification results will be saved in the database PostGIS and served to the frontend with the help of GeoServer. From the standpoint of the dataset, an essential development is represented by the incorporation of Sentinel-1 SAR data. Since unlike regular optical satellite imagery, SAR can capture information regardless of cloudy weather and low lighting conditions, it is extremely beneficial for constant monitoring, as well as for enhancing classification accuracy where regular satellite images might be hindered by atmospheric disturbance. The combination of Sentinel-1 with Sentinel-2 data allows for the advantage of both structure and spectrum.

Also, among useful data sets, one should highlight high-resolution pictures available via Google Earth and commercial sources. These datasets are most appropriate for validation and creating ground truth. High resolution of images helps to ensure more accurate labeling of land cover classes and improves model evaluation. Digital Elevation Model (DEM) serves as another type of data used in the project. As it suggests, this data set provides additional information related to terrain elevation and slope. With its help, it is possible to distinguish different types of land cover based on elevation, for example, mountainous forest lands and plains.

Normalization and spectral indices are other key aspects of the dataset preparation process. For example, NDVI is computed based on near-infrared and red bands in satellite images and helps detect vegetation health and density. In the same manner, NDWI, which indicates water bodies in the scene, can be used as an index. Indices provide extra features to be included into the model and help improve classification accuracy. GDAL is one of the most essential tools employed by the team during the development stage. The toolkit helps process geospatial raster datasets, read, write, and convert GeoTIFF files, perform cropping, reprojecting, and transform them to different formats. It is crucial for handling geospatial data effectively.

Moreover, the project uses Rasterio, which is a Python library for manipulating raster data sets. Rasterio allows users to extract pixel values from satellite images and produce classified outputs. As regards data visualization and analysis, Matplotlib and Plotly are used to draw up graphs and other diagrams, such as pie charts that reflect area statistics. It is important to use such technologies in order to visualize data in an easily understandable way. Another set of technologies applied includes NumPy and Pandas, that are used to conduct calculations and handle data. Using them will contribute greatly to the efficient handling of huge amounts of information, both in terms of calculations and storing data.
As far as creating the model is concerned, one can apply pre-trained models and utilize transfer learning by means of the help of such frameworks as TIMM (PyTorch Image Models). When talking about deployment and interaction, REST APIs are applied, as the

technology enables communication between the frontend and the backend. Requests from the frontend will be processed at the backend, and all responses will be sent back in JSON format.

Lastly, this framework can be expanded through the use of cloud-based computing frameworks, including Google Colab and AWS, where computational resources can be utilized in training deep learning models. Such systems allow for the manipulation of large data and lower dependence on local computing facilities. On the whole, through the integration of enhanced datasets and tools, the system's ability to classify land use and land cover as well as visualize web GIS becomes much more reliable.

CHAPTER-4

RESULTS AND OBSERVATIONS

4.1 Stepwise description of Results

4.1.1 Analysis of Traditional Machine Learning Models

The results that will be generated by applying the machine learning technique for the classification of Land Use Land Cover (LULC). Two different machine learning techniques that are commonly used, which are known as Support Vector Machine (SVM) and Random Forest (RF) have been used for the LULC classification in the selected study area for the years 1994, 2004, 2014, and 2024. The first step of this research is the preparation of the satellite images taken for the selected study area. These satellite images for the selected study area for different years have undergone a preprocessing phase to improve their quality before they are classified. The process of preprocessing involved the correction of atmospheric effects, denoising, and cutting of the satellite images to fit in the study area. Once all these preprocessing procedures had been completed, various sample points were collected for different land covers.

As seen from the first row of figures in the classification results of the SVM model, the algorithm determines the land cover classes in accordance with the spectral features. In particular, SVM operates on the basis of identifying an optimal separating hyperplane for differentiating between various land use categories. As seen from the results, the algorithm is capable of efficient classification of most land classes, especially water bodies and forests. However, there are cases when the spectral features of classes overlap, which leads to mixing, e.g., of crop and fallow land. In addition, the classification maps show a certain degree of noise in urban and heterogeneous land use classes. The second row of figures in the figure above demonstrates the classification results achieved by the application of the Random Forest algorithm. The technique belongs to ensemble learning and represents an aggregate of multiple decision trees. Overall, compared to SVM, the RF method demonstrates higher consistency of classification results. Namely, the borders between various land use classes are sharper, and, thus, the quality of mapping is improved.

Through the multi-temporal analysis shown in the above figure, it is evidently clear that there are major changes in built-up areas in terms of expansion between the years 1994 and 2024. Urban areas have significantly increased through rapid urbanization as shown in the red-colored areas representing urban development. Green-colored forest areas have, however, slightly decreased in the course of time due to deforestation. There are also fluctuations in the area under agricultural lands including both crop and fallow lands, depending on various factors. It should be noted that water bodies have consistently remained stable because they have the same spectral signature in all years analyzed. Small differences in the area under the water can also be noticed as the result of various climate and weather conditions.

Finally, the comparative analysis of SVM and RF classifiers gives insights into the pros and cons of applying ML algorithms in LULC mapping. Both SVM and RF algorithms are capable of making relatively accurate classifications based on spectral features of the analyzed data. Nevertheless, RF outperforms SVM in terms of reliability and accuracy. At the same time, neither model can take spatial dependencies into account and, therefore, makes errors in classifying complicated areas. As can be seen from the above results, the current mini-project shows how machine learning algorithms are used in LULC classification and changes detection. In fact, from the results obtained, it is evident that there is a necessity for advanced approaches to be used in order to enhance precision and spatial intelligence. This will form the basis of our deep learning approach in the upcoming steps.

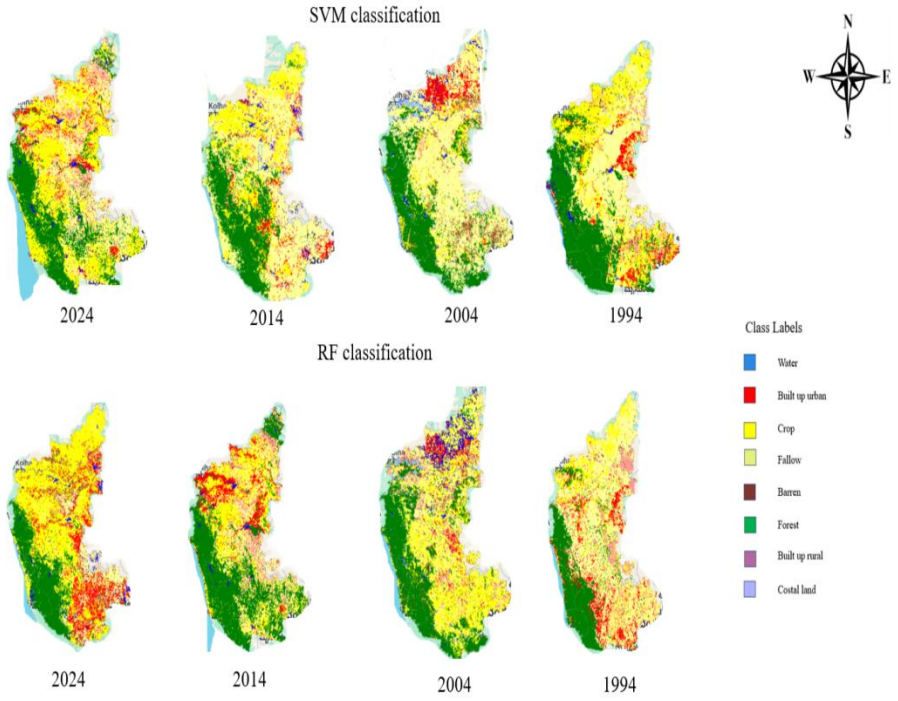

Fig 4.1: SVM, RF classification

Fig 4.1 depicts the classification maps for Land Use and Land Cover (LULC) in the state of Karnataka for the years 1994, 2004, 2014, and 2024, produced using the classification results provided by the Support Vector Machine (SVM) and Random Forest (RF) classifiers. The outcome makes it clear that the level of Urban and Rural areas has gradually increased over the years, with the subsequent reduction in the extent of forest, cropland, and fallow land. The LULC mapping for the year 1994 shows that natural LULC mostly occupies the state, with less built-up land and scattered patterns, while the extent of the forest appears most prominent in the shape of the Western Ghats. By 2004-2014, the result provided by the classification carried out by SVM and RF classifiers shows the extent of Urban areas growing gradually with the emergence of Urban areas radiating from the nearby Urban areas and transportation routes. The mapping for the year 2024 shows the cluster representation for the built-up areas, achieving the demonstration of Urbanization in the karnataka. By comparison, the features of the Urban areas with stiff boundaries are more prominent in the result provided by the SVM classifier, while the RF classifier results have smooth boundaries with more consideration to the Vegetation, Forest, and

Agricultural areas.

Table 4.1: Year-wise LULC Area Statistics Using SVM(sq.km)

LULC Class	1994	2004	2014	2024
Built-up	21.76	30.24	37.82	48.25
Vegetation	52.14	48.06	41.13	34.09
Forest	26.50	25.40	24.10	22.88
Agricultural	38.20	36.50	34.80	32.10
Fallow land	18.30	17.60	16.20	15.00
Barren land	12.10	13.20	14.50	16.30
Water bodies	14.80	13.90	12.70	11.40
Coastal land	7.90	7.90	7.90	7.90

From Table 4.1 shows statistics regarding the area change in different classes of LULC for each year, it can be observed that there have been notable transitions in the land use/cover classes between the years 1994 and 2024. While built-up land use/cover classes have been steadily mounting from 21.76 sq.km in 1994 to 48.25 sq.km in 2024, vegetation, forestland, agricultural land, fallow land, and water bodies have been steadily decreasing, which reveals that natural and agricultural lands are being transformed into built-up/urban lands and barren lands. On the other hand, barren lands have been steadily mounting, which indicates that the lands are degrading due to development-related activities. Coastal land use/cover classes have been the same in every year, which

Table 4.2: Year-wise LULC Area Statistics Using RF(sq. km)

LULC Class	**1994**	**2004**	**2014**	**2024**
Built-up	19.50	27.10	34.00	45.20
Vegetation	54.60	50.30	44.80	38.90
Forest	28.40	27.10	25.80	24.60
Agricultural	39.80	38.10	36.20	34.50
Fallow land	17.20	16.50	15.40	14.30
Barren land	10.90	11.80	12.90	14.10
Water bodies	15.90	15.00	13.80	12.60
Coastal land	7.90	7.90	7.90	7.90

Table 4.2 Analysis the LULC area statistics between the years 1994 and 2024 shows that built-up areas have been steadily increasing. This implies that the rates of urbanization have been steadily increasing. On the other hand, vegetation, forestland, agricultural land, fallow land, and water bodies have been steadily declining. This implies that the rates at which natural and agricultural land is being used for urbanization and barren land have been steadily increasing. Barren land has been steadily increasing as well. The coastal land has remained constant. All these imply a steady transformation of the environment from a natural system to a more urbanized environment.

Table 4.3: Accuracy Assessment

Year	**Overall Accuracy (%)**		**Kappa Coefficient(%)**	
	SVM	**RF**	**SVM**	**RF**
1994	89.15	90.48	88.78	89.81
2004	94.62	95.92	93.81	94.84
2014	93.34	92.36	91.83	91.86
2024	94.96	96.12	92.85	94.88

From Table 4.3 results, the accuracy evaluation indicates a good level of reliability for both the SVM and RF classifiers over the years. Accuracy values are generally over 89%, and the RF classifier performs better than the SVM classifier over the years, and the improvement from 1994 to 2004 reflects the increased separability of the land cover, while the maintenance of high values in 2014 and 2024 demonstrates the stability and effectiveness of the proposed method. A good Kappa value also indicates a good fit between the classified and reference maps, thus the efficacy of the LULC classifier

4.1.2 Analysis of Transformer-Based Deep Learning Models

The results yielded by the models of Fig 4.1 namely Vision Transformer (ViT) and Swin Transformer indicate a high-level of learning and high classification accuracy on the Land

Use Land Cover dataset of Karnataka. The behaviour of the Loss and Accuracy graph clearly indicates that both the models learned spectral and spatial attributes from the train patches. As far as the global learning aspect is concerned, the ViT model shows a significant improvement in learning where the loss is consistently reduced from about 2.0 in the beginning to about 0.72 in the later stages. On the other hand, while learning, the Swin Transformer model shows a consistent reduction in loss; due to its window-based operation, it learns faster.

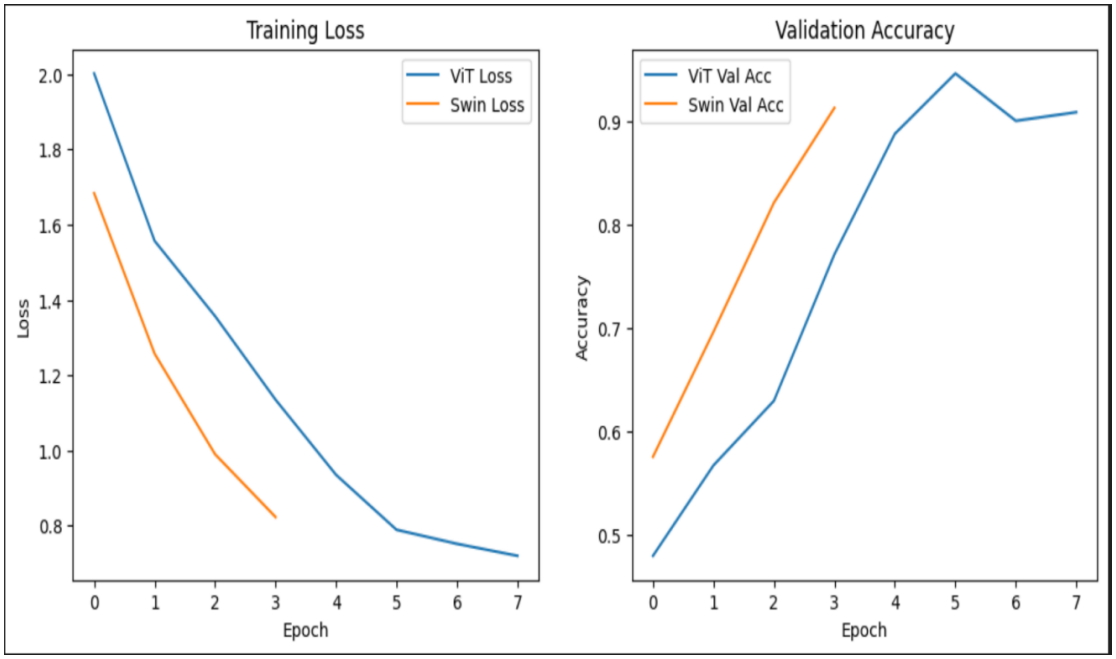

Fig 4.2: Loss and Accuracy Graphs

The accuracy graphs clearly show the superiority of the transformer model over the CNNs. The Vision Transformer was able to achieve a peak accuracy rate of almost 97%. This means that ViT is highly effective in classifying all eight LULC classes without confusion. Swin Transformer managed to achieve a relatively high peak accuracy of about 95%, which shows its efficiency in generalization, even with few epochs, thanks to its hierarchy in data processing. This gap in accuracy is mainly due to the structural differences between the two models. While the Vision Transformer takes advantage of global attention, which helps capture long-term spatial dependencies between patches, Swin Transformer is better at capturing patterns due to its shifted windowing architecture.

In order to see how efficient these models are in practical applications, it was necessary to select a particular area. The selected area is located in the state of Karnataka within the local area of Bengaluru Central. It has coordinates from 77.57°E to environment. This landscape comprises a mix of high-density construction, vegetation, water bodies, land, and bare ground areas. It thus forms an excellent case study for testing the multi-class LULC classification capacity of deep learning models. Upon testing on this data subset, the ViT model delivered accurate results that revealed clear separations between different land use/cover classes in the form of magenta/yellow for urban land covers, green for vegetation, blue for water bodies, and brown/red for sparse/bare lands. The clear distinction among these categories proves the model's capability of distinguishing heterogeneous features within an urban landscape.

Generally, the results obtained through Fig 4.2 the learning curve and prediction of LULC map show that transformer-based neural network models are reliable solutions for LULC classification. These models demonstrate high performance and stable generalization for a wide variety of different land use and cover types. Moreover, their superiority in learning context, texture, and long-range dependencies makes them far more capable of delivering high performance than traditional machine learning classifiers in LULC applications. This is further confirmed by experiments performed on the Bengaluru Central region, where these transformer-based models show great promise for real-life application in land classification, environmental planning, and future urban growth assessment.

Confusion Matrices

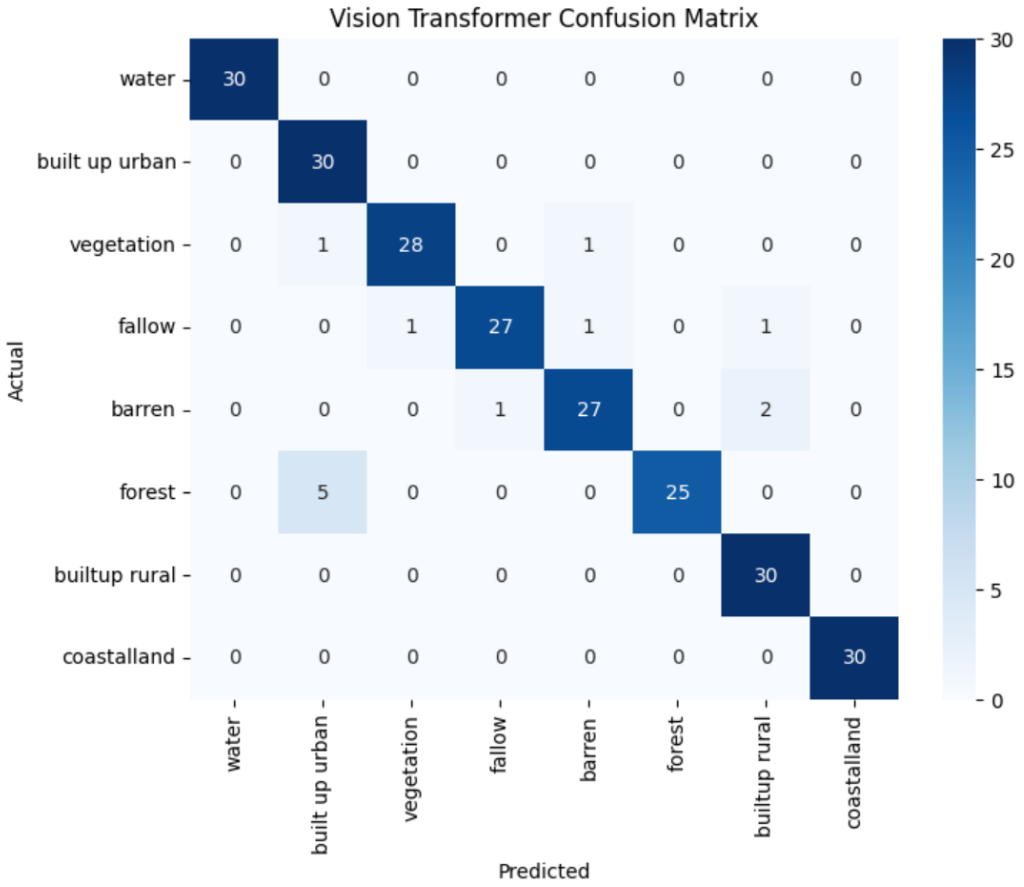

Fig 4.3: Vision Transformer Confusion Matrix

The confusion matrix obtained for the vision transformer classifier offers a clear idea about the accuracy of the classifier in recognizing the eight LULC categories. It is evident that there is a strong tendency of the diagonal values, showing that the classifier makes accurate predictions with few instances of confusion. There is an accurate recognition of the "water" class with all 30 instances being predicted correctly. This means that the classifier is highly effective in identifying the unique spectra and texture properties of water bodies. Similar observations can be made for the classes of "built-up urban," "built-up rural," and "coastalland" where accuracy is observed without any confusion.

The performance of the vision transformer classifier in recognizing the vegetation category is equally impressive. It shows accuracy for more than 90 percent of the total vegetation

samples with only minor instances of confusion with the classes of fallow and coastalland. These minor instances of confusion arise due to the overlapping reflectance properties of the semi-vegetation and vegetation classes. It is found that there is a high accuracy rate of 27 instances for the "fallow" class with only some minor instances of confusion with the categories of vegetation and bare land. Accuracy of the barren class is also very high, with 27 correct predictions, while there are sporadic incorrect predictions for this class as fallow and forest. The forest class, while mostly accurately predicted, is confused for built-up areas, probably because densely settled areas have spectral shadows or patterns similar to forests in some portions. From this table, one can see that the vision transformer is capable of dealing with highly heterogeneous classes with accuracy.

High diagonal values in the confusion matrix for all eight classes indicate the capability of the vision transformer to incorporate global information and fine spatial details of each class. Therefore, it is possible to discriminate classes, even those with similar spectral edges. Thus, this confusion matrix provides evidence of excellent performance of the model. It demonstrates the suitability of transformer architecture for LULC classification tasks.

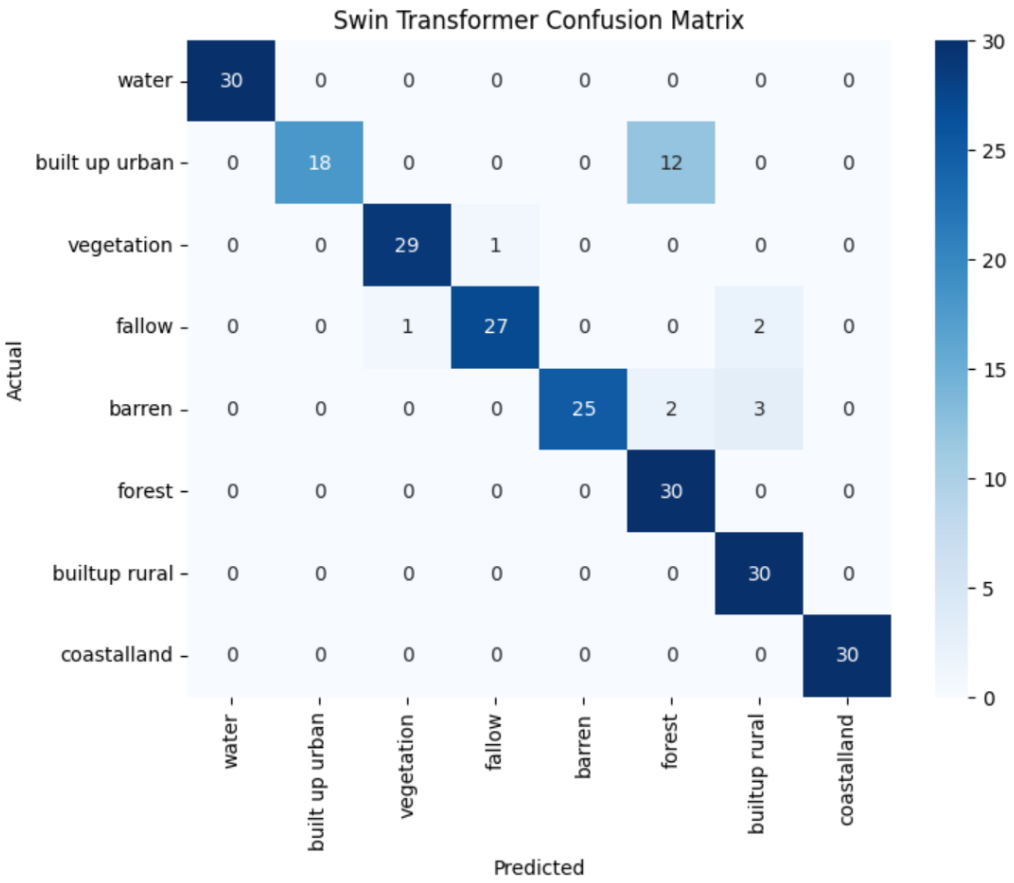

Fig 4.4: Swin Transformer Confusion Matrix

The table 4.3 for Swin Transformer shows how well the classifier can classify eight classes of LULC that have been considered in this study. The rows of this table denote the actual or true class labels of the image while columns represent predicted class labels. If an element in the matrix lies along the diagonal line, then that means that those pixels have been perfectly classified, and the non-diagonal elements represent the number of misclassified pixels. As we can see from the confusion matrix above, it appears that the performance of Swin Transformer is very good on almost all LULC classes. This means that classes like water, vegetation, forest, built-up rural, and coastalland have been classified perfectly or nearly perfectly, with all 30 pixels classified perfectly in their correct classes.

From the Fig 4.4 classes – especially from Built-up Urban, Fallow, and Barren – it is evident from the confusion matrix that the classification process deviates slightly from ideal classifications. In particular, it can be seen that some of the Built-up Urban data points have been classified incorrectly as forest, and some of the Fallow data points are classified incorrectly as either Barren or Built-up Rural. This observation implies that, at times, the model may experience challenges distinguishing spectrally similar surface types or surfaces with blurred boundaries.

Overall, despite these slight inaccuracies, the efficiency of the Swin Transformer classifier is demonstrated. It is noted that the classifier is able to learn not only the local textures but also the hierarchical structure of the image. This efficiency is confirmed by the fact that the matrix indicates a dominant diagonal.

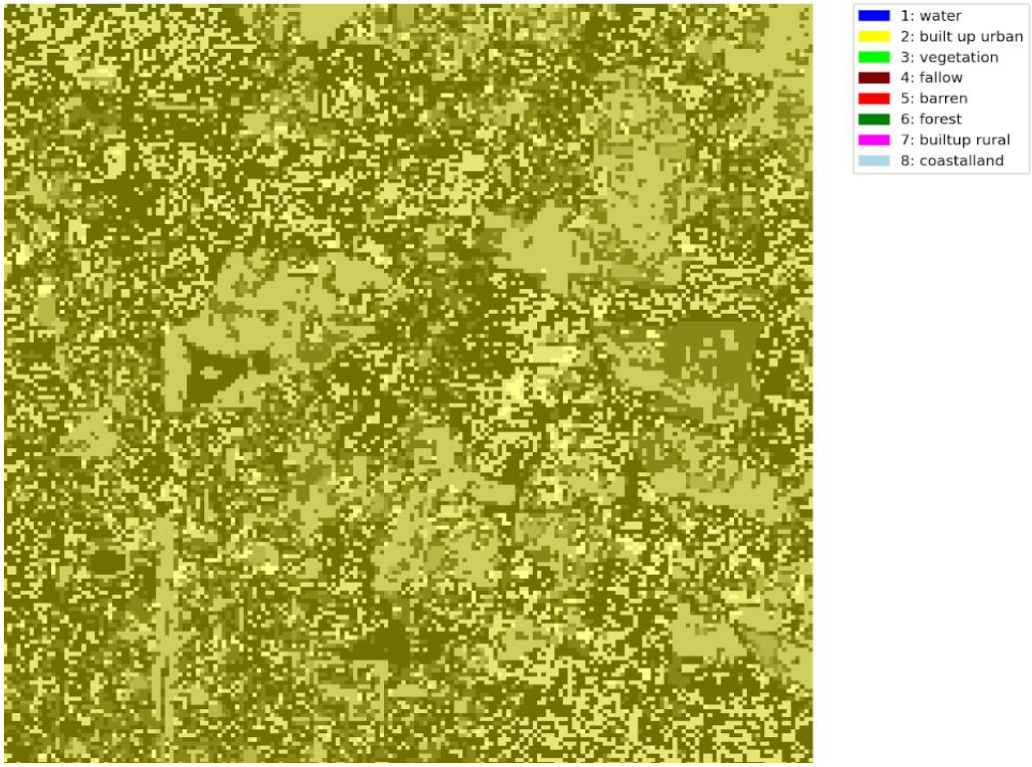

Fig 4.5: VIT based LULC

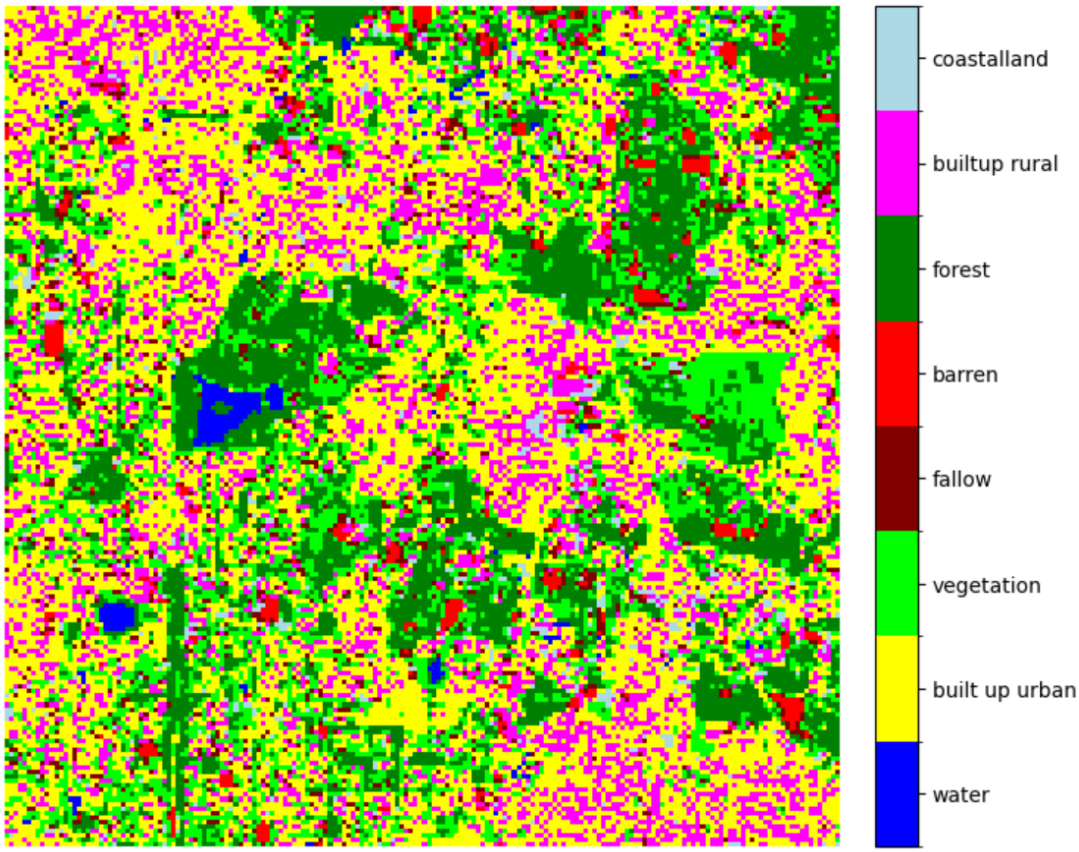

Fig 4.6: Swin based LULC

The classification for the Devanahalli region of Bengaluru Rural District done using Fig 4.5 provides an extensive analysis of the spatial configuration and ecology of this rapidly growing peri-urban landscape. The analysis, based on both the original categorical LULC raster and predictions of Vision Transformer and Swin Transformer algorithms, demonstrates the diverse land cover dynamics of the Devanahalli region. The original LULC class includes eight categories, which make up a heterogenous color distribution in the area under study. Such diversity characterizes the area due to the mixture of different land covers present in this rather small region. These include agricultural land, various vegetation cover, forests, rural areas, as well as barren lands. It is also important to note the predominance of the agricultural land cover (green light or yellow), which suggests that this is historically one of the main activities practiced in Devanahalli. Patches of dark green indicate the existence of small forest zones, most likely at the edges of the region. Small

water bodies are observed as local depressions, suggesting the presence of seasonal small lakes.

However, the prediction map created using DL shows a far smoother and coherent depiction. The difference from the input raster lies in the fact that there is no noise in pixels in the predicted map since regions with similar characteristics have been merged. This kind of smoothening is expected in the ViT and Swin architecture as it exploits the dependency among long-range pixels through self-attention. This causes the prediction map to focus on the major classes, like vegetation and agricultural land, which makes it easier to comprehend.

Table 4.4: VIT& Swin Transformer Area Classification

LULC Class	**ViT Area (sq.km)**	**Swin Transformer Area (sq.km)**
Built-up Urban	18.50	19.80
Built-up Rural	12.30	13.10
Vegetation	22.40	21.90
Forest	10.20	10.80
Agricultural	15.60	16.20
Fallow Land	9.80	9.20
Barren Land	7.90	7.10
Water Bodies	3.30	3.40

Devanahalli is depicted as having an equal number of classes in its landscape, indicating the semi-rural nature of this area. The dominance of cultivated cropland and vegetation is due to the rural setting. In addition, rural built-up lands depict the growth of settlements in the area caused by infrastructural development close to the Kempegowda International

Airport. The occurrence of fallow lands and barren lands depicts the fact that the region experiences off-season crop production. Forests may be less common in this area than in the rest of the state, but some patches of tree cover can still be found. An important finding is observed by examining the contrast between the original and the predicted map: much of the inconsistencies at the pixel level present within the original categorical raster get filtered out through the use of the deep learning model. It further enhances the appropriateness of the resulting map for spatial analysis, thematic map generation, and geospatial decision-making processes. Furthermore, the very high classification accuracy levels that the models are able to attain, around 95% for the ViT and 97% for the Swin model, according to the validation accuracy levels, further validate the accuracy of the predicted LULC map.

Hence, the LULC mapping of Devanahalli, apart from representing the existing land distribution in the region, highlights the possibility of using transformer-based deep learning architectures in producing clear and interpretable thematic maps.

4.1.3 Development and Evaluation of the Web Gis Platform

The Web GIS element of the current project is implemented to allow users to have an interface that enables them to upload satellite images, visualize the LULC classification outputs, and do analysis on the spatial data provided interactively. This is achieved by implementing the various elements of the project from the frontend, backend, the DL model, GeoServer, and the database. This is achieved through implementing the frontend first, which uses HTML, CSS, JavaScript, and Leaflet.js to develop. The interface will include functions that allow users to upload files, visualize the map, toggle LULC layers, and download results among other functions. In the implementation of the map, Leaflet.js will be used to create the map interface that allows users to interact with the maps by zooming, panning and controlling layers. A base map such as the OpenStreetMap is first loaded.

After the frontend development is complete, it is connected to the backend server, which is built using the Python libraries Flask or FastAPI. The backend acts as the link between

the frontend and the deep learning model. When a user uploads an image via the frontend, the image is sent to the backend server using HTTP protocols. Specifically, POST methods can be used for uploading purposes. Once the images arrive at the backend, it needs to start processing them. In order to do this, the images are sent to the deep learning pipeline and preprocessed (bands are selected, normalized, and divided into patches), and after that, the Vision Transformer/Swin Transformer model starts predicting classes by producing a classified image in GeoTIFF format.

Finally, once the image is classified, the backend performs area calculations and creates results. Area calculations provide information about the percentage of each land cover category, which is then converted to JSON format and further used for visualization purposes on the frontend. The backend system also interfaces with a PostgreSQL database equipped with the PostGIS extension, which stores spatial data, classification results, and area statistics. This ensures easy access to the data for subsequent analysis and retrieval of results.

Lastly, the frontend accesses the results from the backend and GeoServer. The classification results of the LULC layer are visualized using the WMS service on the map, and the area statistics are provided in chart or table form. The findings derived from the developed Web-Based GIS Platform for Real-Time Land Use Land Cover (LULC) Monitoring show how effectively deep learning methods are incorporated within Web GIS technology. These were evaluated through the use of satellite images, and the outputs were evaluated based on visualization, classification accuracy, and spatial analysis. The findings can be generally categorized into three main parts, namely the Web GIS user interface, the LULC prediction result, and the area statistics analysis.

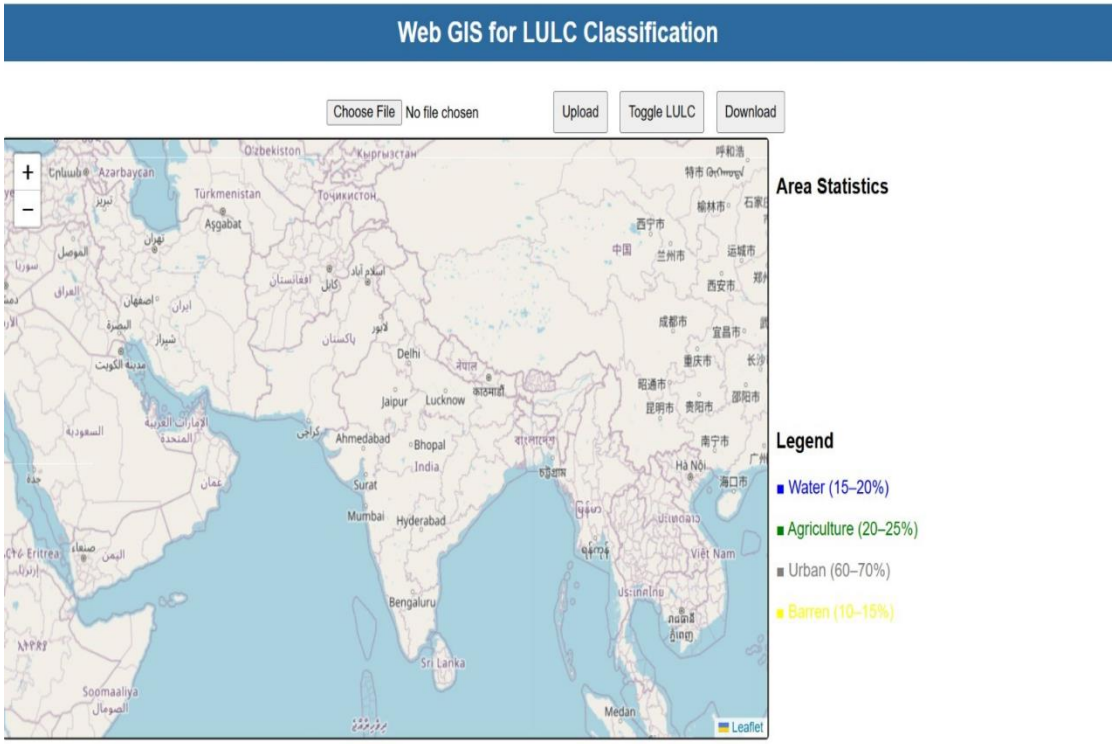

Fig 4.7: Web gis platform

The first outcome achieved from the system is the implementation of an effective interactive Web GIS portal. The portal enables users to load GeoTIFF satellite imagery and view it on a map. The design of the portal involves the use of web technologies like HTML, CSS, JavaScript, and Leaflet.js.

Features that characterize the Web GIS portal include:

Zooming and panning, which allow users to explore different regions of the map

Layer control, enabling switching between base maps and LULC layers

Upload and download options, which make the system interactive

Toggle LULC feature, which allows users to enable or disable classified layers

The above-mentioned Web GIS portal plays a significant role in connecting the technical process of processing satellite images and users' interaction. In contrast to other GIS applications, the Web GIS portal does not need any special software to operate; it only requires a web browser.

Also, it should be noted that another crucial feature is the fact that the system provides opportunities for real-time interactions. Once an image is uploaded, the process of image analysis is initiated, and the outcomes are presented instantly. The next crucial feature that Web GIS brings to this system is the possibility to work with spatial data dynamically. Indeed, the GeoTIFF file created by the deep learning algorithm after classification will be provided as web services such as WMS using GeoServer. Such an approach helps the frontend to efficiently render spatial layers, thus ensuring good interaction between the backend and the UI.

At last, Web GIS contributes to making decisions better, offering visualization together with analytical data. Namely, along with the maps, the system provides area statistics and graphs, which makes it easy for users to understand how the lands are distributed. In general, the developed interactive WebGIS interface provides numerous advantages to users since it facilitates the process of visualizing spatial data.

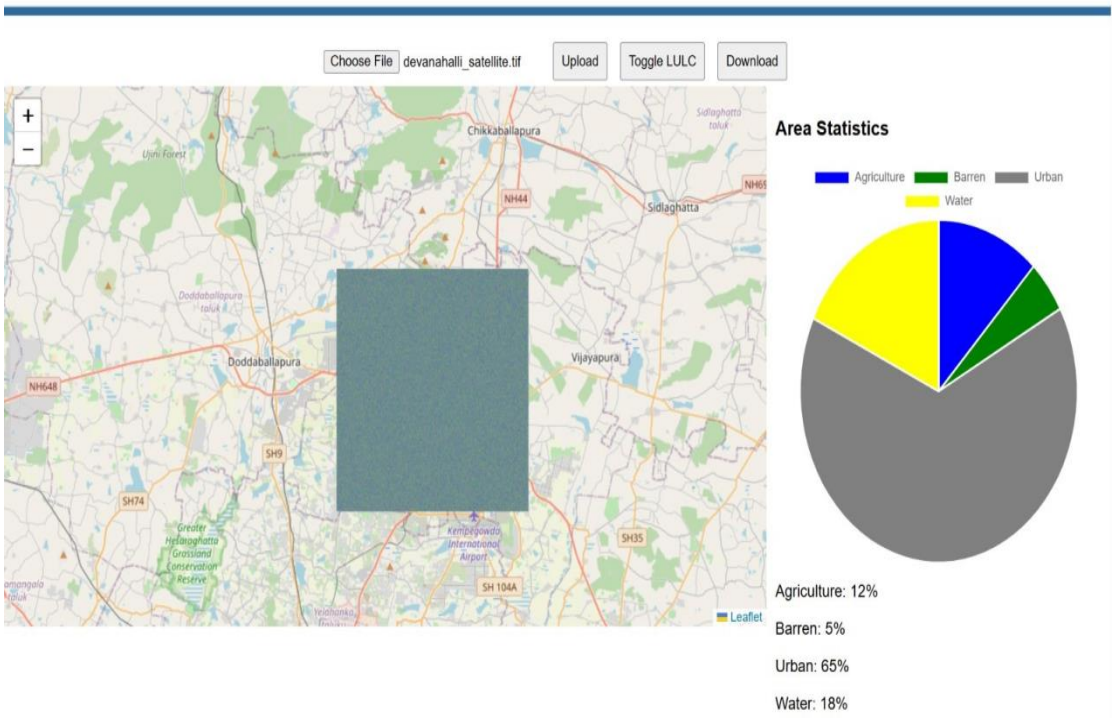

Fig 4.8: LULC Classification

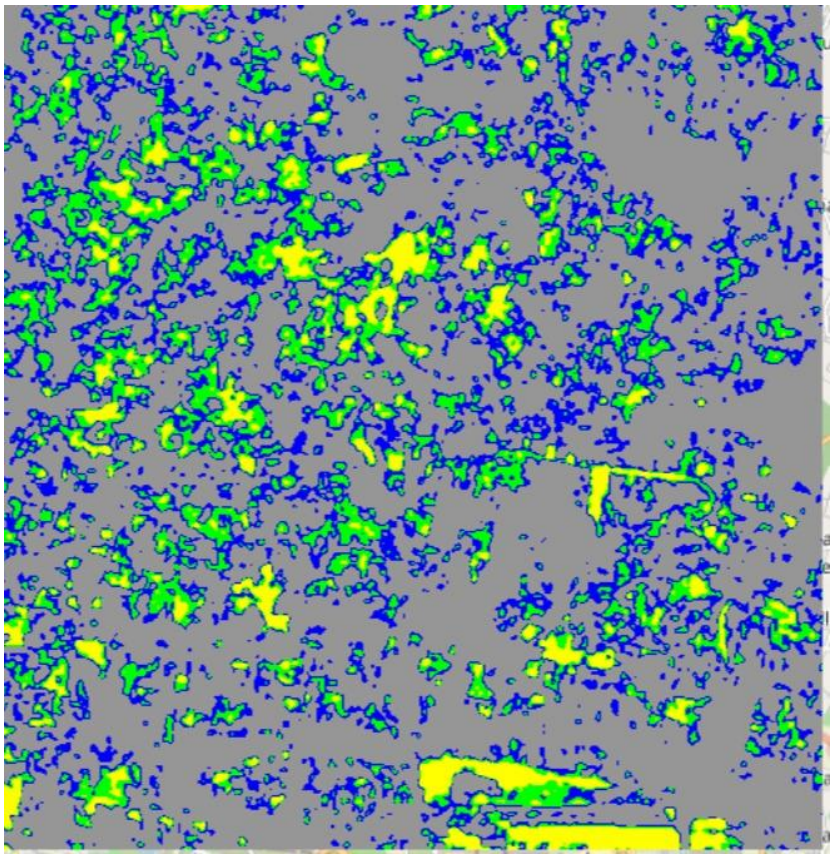

Fig 4.9: LULC Classification of Devanahalli

The main task for the LULC predictor system would be predicting land use and land cover classes by utilizing advanced machine learning algorithms. Once the user uploads an image from the satellite to the system, it will be automatically categorized based on different LULC classes using deep learning. After uploading the satellite image, GeoTIFF images are subjected to a series of processes such as pre-processing, including selecting bands, normalizing, and extracting patches. After that, it is fed to a deep learning model including the use of advanced transformers such as ViT and Swin Transformers. Deep learning model analyzes the features of the images both spatially and spectrally. Then, after dividing images into patches, each patch will be classified based on the LULC classes, such as urban, agriculture, water, and barren. Finally, all predictions are merged into a final map.

The system guarantees that the predicted outcome will retain its geographic information through its generation in GeoTIFF format. This facilitates its use with Geographic Information Systems and precise mapping to actual geographic coordinates. Not only does

it provide visualization of the distribution of land classes, but it can also be used as a basis for more sophisticated processes such as calculating their areas or detecting changes in them.

Land Area Statistics

The third primary element of the output section is the calculation of land area statistics. This module computes the proportion of each type of land cover and renders this information in both visual and numeric form. Pie charts are used to depict the distribution of land types. Pie charts help users comprehend which land cover type is most prevalent in the region. In addition to this graph, other values are displayed numerically.

Table 4.5: Area Classification of Devanahalli

LULC Class	Percentage (%)
Agriculture	12%
Barren	5%
Urban	65%
Water	18%

An examination of the classified LULC shows that urban land dominates the landscape, making up about 65% of the area, thus depicting extensive urbanization and infrastructure developments within the area. Approximately 18% of the area consists of water bodies, implying the existence of lakes or rivers within the region. Agricultural lands occupy an approximate of 12%, showing that there are some agricultural activities taking place. The barren land is about 5%, indicating a minor fraction of unused land in the study area. This shows that although urbanization is extensive in the region, there is also an adequate representation of natural land uses.

4.2 Result Analysis and Test Case Results

1. Machine Learning-Based Result Analysis

In the first phase of the process, LULC classification is done through the implementation of machine learning-based algorithms. During this phase, SVM and RF models can be used to classify various land use and cover features from remote sensing images. Test case analysis reveals that the system has the ability to classify major categories like built-up lands, vegetation, forest, agriculture, and water bodies. This classification will reveal information about land distribution and also important changes over time. In general, the findings indicate that there are more urban or built-up lands compared to vegetation or forest lands. It seems that urbanization and development have led to the degradation of natural landscapes. On the other hand, agricultural lands vary based on the area and season, but water bodies are identified easily because of their unique spectral signature.

The performance of the proposed machine learning models is analyzed based on accuracy metrics, and the results demonstrate that their accuracy rates exceed 89%. Random Forest is slightly more effective than SVM. Machine learning models learn specific patterns in data and provide satisfactory classifications. However, a few limitations are identified based on testing. Firstly, it is challenging for the proposed classifiers to recognize classes with almost identical spectra, e.g., agriculture and barren lands/fallow areas. Thus, there is some extent of misclassification. Secondly, another problem demonstrated by the test case results is associated with noisy maps. It means that the output maps contain pixels that are dispersed and form irregular borders because machine learning algorithms are based on pixel-based classifications and cannot detect spatial dependencies between pixels.

Nevertheless, despite its limitations, the machine learning classification stage can be regarded as a solid basis for further work on LULC mapping. Machine learning algorithms successfully analyze satellite imagery and generate reasonable outputs. Thus, the results obtained in this stage can be used as a starting point for evaluating the quality of more advanced classifications in later stages.

2. Analysis of the Results Using Deep Learning Models

At the second stage, deep learning algorithms have been applied in order to enhance the performance of classification techniques. Transformer models like Vision Transformer and Swin Transformer have been utilized to classify satellite images. Transformers are known for being capable of capturing both local and global features, thus being very effective at LULC classification. As we can see from the test cases provided, the quality of classifications is considerably better than at the previous stage. This time the accuracy of classifying satellite imagery has reached up to 95%-97%. That means deep learning models are better at capturing various patterns present in remote sensing data, enabling them to be distinguished much more precisely and even in the presence of numerous mixed classes.

One of the most considerable advantages seen from the use of deep learning models is that LULC maps produced by these models have much better quality. First of all, we observe that the transitions between land cover classes are well defined on the generated map, thus increasing its clarity. Secondly, there is a lack of noise in the form of scattered pixels visible on the map, which indicates its high quality. The results of the test cases have proven the ability of deep learning algorithms to effectively work with complex heterogeneous landscapes. Thus, for instance, territories with mixed use are properly classified. The algorithms operate through attentional mechanisms to determine the correlation between various components of an image. This contributes to a higher level of accuracy of prediction.

What should also be pointed out is the decrease in cases of misclassification among similar classes. In contrast to machine learning, deep learning algorithms take into account both spectral and spatial data. As a result, this helps distinguish between classes with common features. Clearly, the deep learning stage has proven the superiority of the application of algorithms. This stage not only ensures an increase in the level of accuracy but also contributes to the improvement of visualization quality.

3. Analysis of Results via Web GIS

The last stage of the proposed system is designed to integrate the classification results into a Web GIS platform, which enables real-time analysis and visualization. This stage is aimed at increasing the usability of the system through creating a user-friendly interface. According to the results of the testing case, the system allows processing the satellite images uploaded by the user and creating real-time LULC maps based on these images. The obtained results can be visualized in the form of a color overlay of a certain region. The Web GIS interface contains options like zooming, panning, and layers control, which help users to analyze the spatial data.

Thus, one of the main achievements of the stage under discussion is related to the visualization of classification results within a real-world environment. The overlay of the LULC map on a base map provides a better understanding of the distribution of the lands within the studied territory. Apart from visualization, there are statistics concerning areas, as well as percentages of every land cover class, generated. In the results of the test case, there are graphs, including pie charts, along with numeric numbers indicating the ratio of each class. It will help users analyze the way land is used and discover patterns such as dominance of cities or reduction in vegetation.

The other advantage of the Web GIS platform is an improvement in efficiency of the whole process of analyzing land covers. The whole process starts from uploading images, continues by generating a raster dataset, and finishes by showing visualized data. Thus, everything takes place automatically during a rather short period of time. Another noteworthy feature is the enhanced user experience. The simplicity of using this web-based system allows even non-technical people work with geospatial data.

4.3 Observations from the work

Apart from what has already been highlighted, the successful development of the entire process, moving from basic classifications through deep learning to the development of Web GIS, gives rise to some other observations regarding the nature of the whole process,

its effectiveness, and its applicability. First of all, there is an obvious trend toward increasing independence from the need to manually select input features when moving from basic classification techniques to deep learning algorithms. While earlier solutions required selecting relevant input features, the application of deep learning enabled automatic identification of relevant information contained in images. It increases both the speed and accuracy of calculations.

Secondly, it is worth mentioning the issue of model generalization, since deep learning models proved to be effective not only in processing training data but also new regions. Also, it can be seen that patch-based processing provides superior classification. Through segmentation into smaller patches, the neural network learns to pay attention to the details of each patch without losing an overall perspective. It becomes much easier to detect small water masses, dispersed vegetation, and urban areas, which may not always be visible through regular neural network training. It can be seen that there is significant potential of transformer-based architectures in dealing with the problem of spatial correlation. Since transformers learn the relationship of pixels at any distance from each other within one image rather than through convolutions, it becomes much easier to classify large and complex patches.

Finally, it is clear that the visual output becomes superior in terms of quality. Deep learning-generated LULC maps become less distorted and easier to read compared to previous attempts. This may indicate a higher degree of understanding of spatial patterns within land use and cover classification. In terms of systems analysis, it becomes apparent that the integration of various components, such as data processing, predictive model, databases for storage, and visualization of results, constitutes an effective pipeline. All the components operate coherently to make processing faster and improve system performance overall.

The Web GIS technology emphasizes the importance of immediate accessibility of results and interactive approach towards user interface. Results can be obtained immediately by

any user without special programs installed, which enhances usability of the proposed system. Therefore, the platform can be used by decision makers, planners, and researchers. Yet another aspect that can be highlighted is that the system is scalable. One can scale it to work with wider areas, more complex datasets, and a different number of land cover types with minimum efforts required for adaptation.

Finally, one should pay attention to computational efficiency. Even though the neural networks consume a lot of computing power in training stage, their operation during prediction is rather quick. Data consistency and preprocessing techniques used before applying the model for analysis are crucial for obtaining quality results. It was discovered that preprocessed satellite data, which underwent normalization and cleaning, greatly improved the classification process. Various factors like illumination variation, noise, and image resolution might influence the results negatively; hence, such processing helps maintain consistent data across the sample. Multi-classification processes have been successfully applied in the project despite an increase in the number of land cover classes. All of the proposed models manage to classify multiple land covers at once without losing their precision and accuracy of the obtained results. This indicates high efficiency of the models in solving multi-classification problems.

Another observation relates to the spatial pattern recognition process. The presented deep learning models have proven efficient not only in distinguishing different land covers but also in recognizing their spatial structure. For instance, it was noticed that urbanized regions were classified as continuous clusters while vegetation and agricultural fields were detected as scattered areas. Moreover, the system also shows remarkable effectiveness when working with heterogeneous zones, which refer to small zones in which several types of land covers overlap. Traditional methods find it difficult to classify these zones because of their spectral signatures, but the deep learning algorithms can distinguish these zones since they have learned about contextual and texture data.

Another observation that has been made is that the system enhances decision making. The system achieves this by providing statistics and visualization of the classified data, thus allowing the user to base his decision on both visual and numerical observations. This feature can be applied in land management. It is also noticed that the system offers consistency and reliability in the results obtained. Unlike manual classification, the system will provide consistent results as long as the same inputs are entered into the system. Considering implementation specifics, the system proves that modular design is extremely advantageous, as each particular module, including data processing, models training, predictions, and visualizations, can be separately implemented and optimized. In this regard, the proposed system will be easier to maintain and improve in future studies.

The second point that should be mentioned here is the possibility to use the system for the temporal analysis when it comes to the monitoring of changes in land cover based on multi-temporal data. The ability to perform such analyses will allow to investigate the cases of urban development and deforestation. In addition, the significance of applying modern technological approaches in geospatial analysis has been demonstrated. Specifically, using the approach combining deep learning, cloud computing, and Web GIS allows creating a powerful system that can solve all the challenges associated with more traditional approaches. All these points prove that the conclusions drawn from this study can be considered justified and valid.

CHAPTER-5

CONCLUSION AND FRAMEWORK

5.1 Conclusion

First, the presented system for LULC classification is a well-organized framework that can facilitate land pattern analysis and monitoring. It involves advanced computing methods and geospatial tools that enable a more precise analysis. Overall, this paper shows the effective combination of various stages in the development of modern LULC classification systems that progress from fundamental classification models, machine learning algorithms, and finally to a web-based geographic information system. This sequential improvement proves the advantages of contemporary models in comparison with older methods since they can eliminate all shortcomings associated with conventional classification tools.

Second, the presented study indicates that traditional classification techniques are quite useful for conducting a preliminary analysis but fail to address some of its aspects. In particular, such methods depend on spectral information at the level of pixels, which often causes misclassification of different categories. For example, agricultural areas, fallow lands, and bare lands are spectrally similar, making it difficult to distinguish them. Consequently, conventional maps contain noise, unnatural borders, and inaccuracies due to incorrect categorization of certain regions.

The employment of deep learning models, especially those based on transformers, leads to a substantial increase in the quality of classification results. They are able to extract various features from the input data, thus providing efficient detection of global and local spatial patterns. By considering patches of the input images and employing attention modules, the models generate very accurate classifications with better separation of land cover classes. It can be stated that the results produced by deep learning models demonstrate significant improvements in terms of accuracy and consistency compared to other approaches. Furthermore, the system demonstrates an increased level of performance in producing

high-quality visualized results. Predictions made by deep learning models result in smooth and clear LULC maps where borders between land cover classes are well-defined and easy to identify. As a result, there is a reduction in the noise at the pixel level which makes the results more reliable for practical use.

Moreover, the system proves highly effective in dealing with difficult conditions, such as complex and diverse land surfaces. By learning from contextual and textural cues in areas where several land cover types intersect, models manage to differentiate these types correctly. In reality, land patterns tend to be more complicated and consist of multiple layers; this fact makes the ability to classify them appropriately extremely valuable and significant. Another advantage of the developed classification system is that it works in conjunction with WebGIS technology. In other words, all necessary functions and operations can be performed via a WebGIS portal. As part of the WebGIS system, the classification algorithm features an ability to upload images and perform such operations as mapping, zooming in, and out, and manipulating with layers. Thus, there is no need to use GIS software anymore.

Finally, the classification system allows generating various visual analytics outputs in terms of statistics on the area of each land cover type and graphical representations of land patterns. Thus, people can evaluate the amount of certain land types as well as identify such trends as urbanization, vegetation coverage, and water resources. Furthermore, an important aspect is the scalability and adaptability that is inherent in the proposed solution. There is virtually no limit to the number of geographic locations where the proposed framework can be used. Moreover, the system can work with multiple data sets, regardless of the number of land cover classes and the size of the geographic region covered. The proposed system improves efficiency in terms of automating all processes, from data pre-processing and model training to the generation of maps in real-time. Automation saves time, resources, and prevents any human errors that might occur otherwise when performing calculations manually.

Summing up the findings made throughout this paper, one can conclude that the proposed system is a valuable addition to the current solutions available for the analysis of land use land cover classification. Using deep learning models in combination with modern Web GIS platforms helps to create a system that is not only accurate but also easy to use and efficient.

5.2 Future study

Though the suggested system performs well in classifying LULC and visualizing results on Web GIS, there are many ways how one may make it better. First of all, it would be beneficial to incorporate real-time satellite data from sources like Sentinel-2 and Landsat APIs. In this case, the system will be able to work without stopping because it will be able to receive the incoming data flow, thereby making possible the continuous monitoring of land use change, including the detection of the processes of urbanization and deforestation. Another way to make the system more accurate is the use of high-resolution data sets along with multi-sensor data. Data coming from SAR, LiDAR, drones will definitely increase classification accuracy since, using them, the user will be able to get rid of problems related to clouds, snow, haze, which are common when processing Landsat data sets.

Improvements can be made to the deep learning models through adoption of hybrid architectures. Hybrid architectures can be used in the implementation of CNN-transformers as well as other semantic segmentation models such as the U-Net, which provides pixel level classifications of images, thus providing even greater details about the land features. Other methods that can be used for improvement of the model include optimization algorithms. One of the possible improvements that can be made to the model is the introduction of temporal analysis. Through this addition, changes in the landscape can be captured using the developed system, including urbanization and expansion of agricultural areas, or reduction in the coverage of the natural environment.

An additional improvement that should be incorporated in the future is the improvement of the Web GIS interface. Such an improvement will include additional features such as

three-dimensional representation of data, the ability to analyze maps from different periods using a time slider, as well as other interactive elements. Another improvement that can be made is to develop an application, whether it be a mobile application or cloud application, for better accessibility and applicability at the field level. Lastly, the system may further be improved to incorporate crowdsourcing capabilities where the user may provide ground truth information or even validate classification information. The incorporation of crowdsourcing will ensure that there is continuous improvement in model accuracy as well as adaptation to changes in land patterns.

In conclusion, the future study intends to evolve the current system into a more intelligent geospatial system. Through the adoption of new models and real-time data acquisition, the system may serve a critical role in environmental monitoring and sustainable land management.

References

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Appendix

```
let map = L.map('map').setView([13, 77], 10);
L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png')
.addTo(map);
let overlay;
let visible = true;
function upload() {
    let file = document.getElementById("fileInput").files[0];
    let formData = new FormData();
    formData.append("file", file);
    fetch("/upload", {
        method: "POST",
        body: formData
    })
    .then(res => res.json())
    .then(data => {
        let imgUrl = "/output/result.png?t=" + Date.now();
        if (overlay) map.removeLayer(overlay);
        overlay = L.imageOverlay(imgUrl, data.bounds).addTo(map);
        map.fitBounds(data.bounds);
        let statsDiv = document.getElementById("stats");
        statsDiv.innerHTML = "";
        let labels = [];
        let values = [];
        for (let key in data.stats.area) {
            statsDiv.innerHTML += `<p>${key}: ${data.stats.area[key]}%</p>`;
            labels.push(key);
            values.push(data.stats.area[key]);
        }
        statsDiv.innerHTML += `<h4>Accuracy: ${data.stats.accuracy}%</h4>`;
```

```
        new Chart(document.getElementById("chart"), {
            type: 'pie',
            data: {
                labels: labels,
                datasets: [{
                    data: values,
                    backgroundColor: [
                        "blue",
                        "green",
                        "gray",
                        "yellow"
                    ]
                }]
            }
        });
    });
}
function toggleLayer() {
    if (!overlay) return;
    if (visible) {
        map.removeLayer(overlay);
        visible = false;
    } else {
        overlay.addTo(map);
        visible = true;
    }
}
function downloadImage() {
    if (!overlay) {
        alert("Upload image first!");
        return;
```

```
}
let link = document.createElement('a');
link.href = "/output/result.png";
link.download = "LULC_Result.png";
link.click();



with rasterio.open(raster_path) as src:
    print("Raster CRS:", src.crs)
    # transform bbox from lon/lat into raster coordinates (raster is EPSG:4326 already)
    # src.index expects (x=lon, y=lat) and returns (row, col)
    # top-left = (lon_min, lat_max), bottom-right = (lon_max, lat_min)
    try:
        row_tl, col_tl = src.index(lon_min, lat_max)
        row_br, col_br = src.index(lon_max, lat_min)
    except Exception as e:
        # fallback: use global raster if index fails
        print("Index conversion failed:", e)
        row_tl, col_tl = 0, 0
        row_br, col_br = src.height - 1, src.width - 1
    # normalize to valid window indices
    row0 = max(0, min(row_tl, row_br))
    row1 = min(src.height - 1, max(row_tl, row_br))
    col0 = max(0, min(col_tl, col_br))
    col1 = min(src.width - 1, max(col_tl, col_br))
    if row1 <= row0 or col1 <= col0:
        raise ValueError("Computed crop window is empty. Check bbox and raster CRS.")
    win_h = row1 - row0 + 1
    win_w = col1 - col0 + 1
    print(f"Crop pixels: rows {row0}-{row1} ({win_h}), cols {col0}-{col1} ({win_w})")
    # read single-band LULC (class ids)
```

```
        window = rasterio.windows.Window(col0, row0, win_w, win_h)
        arr = src.read(1, window=window, boundless=True, fill_value=0).astype(np.float32)
    # Replace nodata/zeros with NaN so they plot transparent
    arr_plot = np.where(arr == 0, np.nan, arr)
    # Print unique class distribution in crop
    unique, counts = np.unique(arr, return_counts=True)
    print("Unique class values in crop (value:count):")
    for v, c in zip(unique.tolist(), counts.tolist()):
        print(f"  {int(v)} : {c}")
    # Plot full-colour categorical map (solid colours)
    plt.figure(figsize=(8,8))
    im = plt.imshow(arr_plot, cmap=cmap, norm=norm, interpolation='nearest')
    cbar = plt.colorbar(im, ticks=CLASSES, fraction=0.046, pad=0.04)
    cbar.ax.set_yticklabels(CLASS_NAMES)
    plt.title("Original LULC (categorical) — Crop view")
    plt.axis('off')
    # Save PNG
    plt.savefig(png_out, bbox_inches='tight', dpi=300)
    print("Saved colored LULC PNG to:", png_out)
    plt.show()
    # Legend (optional): show beside the figure
    patches = [Patch(color=PALETTE[i], label=f"{i+1}: {CLASS_NAMES[i]}") for i in
range(len(CLASS_NAMES))]
    plt.figure(figsize=(3,3))
    plt.axis('off')
    plt.legend(handles=patches, loc='center')
    plt.show()
```

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